Parallel Fast Entangling Gates for Trapped Ion Quantum Computing

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Declaration

This thesis is an account of research undertaken between February 2026 and October 2026 at the Research School of Physics, The Australian National University, Canberra, Australia.

Except where acknowledged in the customary manner, the material presented in this thesis is, to the best of my knowledge, original and has not been submitted in whole or part for a degree at any other university.

Edward Troccoli

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Abstract

Trapped-ion quantum computers have demonstrated the highest fidelity two-qubit entangling gates using highly-connected qubits. However, scalable quantum computation additionally requires fast gates that can be enacted in *parallel*. 'Fast gates' have been theorised as a method for reducing the duration of two-qubit gates by using impulsive state-dependent momentum kicks to simultaneously drive the collective motional modes of the trapped ions. Current 'slow gates' perform parallel gate operations but are limited by crosstalk between ion pairs [1]; the impact of crosstalk on fast gates remains unexplored. This thesis therefore investigates parallelisation of fast gate protocols via a theoretical study of two-qubit entangling gates in 10-ion chains. I first examine the naive parallelisation of independently optimised two-qubit gates, finding that high-fidelity operation is achievable for well-separated nearest-neighbour pairs, whereas adjacent and non-nearest neighbours configurations suffer substantial crosstalk. To overcome this, I develop a novel protocol by extending the current two-qubit framework so that fast gates can be optimised for parallel operation thereby explicitly suppressing crosstalk. I show this protocol suppresses crosstalk for gates of duration 1 µs to 20 µs, improving their infidelity by five orders of magnitude in the case of nearest neighbour pairs. High-fidelity parallel gates involving non-nearest neighbour pairs are possible only for gate times where motional dynamics are non-localised. Comparing gate time and SDK requirements against equivalent serial implementations identifies that non-nearest neighbours parallel gates can approach a twofold speedup for low fidelity thresholds, but that advantage is reduced for high-fidelity gates. Conversely, naive parallelisation remains particularly resource-efficient for well-separated nearest-neighbour pairs, highlighting the difficulty of finding high-fidelity parallel gates as a key limitation. Moreover, I find that laser pulse errors remain a limiting factor for parallel operation and propose alternative methods for realising parallel quantum computation with trapped ions.

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Chapter 1

Introduction

Realising the computational advantages promised by quantum computing requires both algorithms that outperform classical approaches and the ability to execute these algorithms efficiently. By exploiting the uniquely quantum properties of superposition and entanglement, quantum computers can manipulate quantum states allowing the exploration of an exponentially large Hilbert space, providing a natural framework for simulating quantum many-body systems. Beyond quantum simulation, quantum algorithms are useful for computationally demanding problems such as fluid dynamics [2], where convergence of scale-dependent dissipation proves challenging even with 2048^3 grid cells [3].

Despite substantial experimental progress, realising the full potential of quantum computation remains challenging. Quantum information must be encoded in physical quantum bits (qubits) that are susceptible to decoherence, while imperfect control introduces errors during quantum logic operations, known as gates. Quantum error correction provides a pathway towards mitigating these errors (fault-tolerant computation) by encoding logical qubits across multiple physical qubits, but introduces considerable resource overheads [4]. Consequently, the computational capabilities of a quantum processor depend not only on the number of available qubits, but also the connectivity (the extent to which arbitrary qubits can be pairwise entangled), the degree of parallelisability of operations and the speed and accuracy with which they can be performed.

Two-qubit entangling gates, together with single-qubit operations, provide a universal gate set [5] forming the building blocks of any quantum algorithm. Unlike single-qubit gates, entangling operations require interactions between qubits, making their physical implementation particularly demanding. Moreover, the execution time of a quantum algorithm depends not only on the duration of individual gates, but also on the number of operations that must be performed sequentially. Consequently, reducing the duration of individual entangling gates and increasing the number of gates that can be executed in parallel offer two complementary strategies for reducing quantum circuit execution times.

1.1 Leading quantum computation platforms

Quantum computation can, in principle, be realised using any physical system capable of encoding qubits and providing sufficient control to implement quantum logic operations. Several platforms have emerged as leading candidates, including superconducting circuits [6], neutral atoms [7] and trapped ions [8]. DiVincenzo established five criteria for the physical realisation of quantum computation [9]: (i) a scalable system of well-characterised qubits, (ii) reliable qubit initialisation, (iii) coherence times substantially exceeding gate operation times, (iv) a universal set of quantum gates, and (v) qubit-specific measurement capabilities. Although these criteria establish the fundamental requirements of a quantum processor, satisfying them does not necessarily guarantee efficient execution of complex quantum algorithms. In particular, computational performance also depends on the accuracy (fidelity) and duration of quantum gates, the connectivity between qubits, and the extent to which operations can be parallelised.

Quantum circuits consist of sequential layers of gates, with the number of layers defining the circuit depth, D [10]. Assuming comparable gate durations t_g and neglecting additional overheads, circuit execution time scales approximately as Dt_g, motivating both faster individual gates and the parallelisation of otherwise independent gates to reduce circuit depth. However, parallelisation requires sufficient qubit connectivity and control, including the suppression of unwanted interactions, termed *crosstalk*, between simultaneously addressed qubits [1]. Whilst some platforms may exhibit naturally parallel gate operation, this often comes at the cost of poor connectivity and vice-versa. A platform that facilitates both highly connected qubits with highly parallel two-qubit gates is therefore desirable for fault-tolerant quantum computing.

To be considered a viable quantum computing platform, it is necessary to have strong gate performance in terms of both fidelity and gate speed, long coherence times, high-connectivity between qubits and parallelisation of gates. Superconducting circuits support fast, high-fidelity entangling gates and parallel operations between nearest-neighbour qubits. However, as superconducting qubits typically operate via nearest-neighbour interactions, their connectivity is limited requiring additional overheads which slow the overall computation [23]. Another approach, involving neutral-atom arrays combines

Table 1.1: Representative metrics for three of the most performant quantum computing platforms. Reported are the gate duration and fidelities for two-qubit entangling operations, along with the amount of these gates that have been performed in parallel. Qubit coherence times are reported to provide a reference for the gate times, noting that a high coherence time to gate duration ratio is favourable. Importantly, these metrics are from independent studies and do not necessarily imply that they can all be currently simultaneously achieved.

	Two-qubit entangling gates		Coherence time [s]	Parallel 2Q gates demonstrated
Platform	Duration [s]	Fidelity [%]		
Trapped ions	$\sim 4.8 \times 10^{-7}$ [11]	99.99 [12]	$\sim 5.5 \times 10^{3}$ [13]	4 [14]
Superconducting qubits	$\sim 1.2 \times 10^{-8}$ [15]	99.94 [16]	$\sim 1.48 \times 10^{-3}$ [17]	22 [18]
Neutral atoms	$\sim 2.7 \times 10^{-7}$ [19]	99.71 [20]	~ 40 [21]	30 [22]

flexible spatial arrangements with parallel entangling operations mediated by Rydberg interactions, but their performance depends heavily on the fidelity of these interactions. Moreover, moderate connectivity is achieved by using optical tweezers which requires additional processes such as cooling atoms after each shuttling process [22]. Trapped ions offer fast, high-fidelity quantum operations and high connectivity through collective motional modes allowing any two qubits to interact. However, these shared modes can also mediate crosstalk, requiring complicated control of qubits to enact parallel entangling gates [1]. It is therefore clear that simultaneously satisfying the conditions for a viable quantum computing platform is not a straightforward task and is precisely why there exist many approaches to quantum computing. A comparison table of state of the art performance for these platforms is given in Table. 1.1. The following section further explores trapped ions and outlines an approach to decreasing gate times whilst also enabling parallel operation.

1.2 Trapped ions as a platform for quantum computation

Trapped-ion quantum computers encode qubits in the internal electronic states of electromagnetically confined atomic ions. Their mutual Coulomb interactions couple the ions' motion, allowing laser-driven entangling operations between arbitrary ion pairs through

shared collective motional modes. This means that any two ions can be entangled[1] without requiring physical transport of ions, although the shared motion also introduces challenges for independently controlling parallel operations [1].

Trapped-ion entangling gates, such as the Mølmer–Sørensen gate, typically rely on an adiabatic (with respect to the trapping frequency) condition where laser-driven interactions near a single motional sideband generates entanglement [24, 25]. The need to adiabatically control these interactions while suppressing unwanted motional excitation increases gate times, particularly in longer ion chains where the increasing number of motional modes complicates control. Long gate times are undesirable as an insufficient number of gates for quantum algorithms can be completed within the finite qubit coherence time. One approach to improving how trapped-ion quantum computers scale with increasing ions involves distributing ions across multiple trapping zones, enabling intrinsically parallel operations at the expense of having to physically shuttle ions between trapping zones [14, 26]. Whilst shuttling ions enables connectivity, it is a very slow process making up 95% of the total gate time [27]. To push past the scaling bottlenecks of adiabatic trapped-ion entangling gates, alternatives have been proposed to drive many motional sidebands therefore enabling gate times up to two orders of magnitude faster.

1.2.1 Fast entangling gates and the challenge of parallelisation

Impulsive fast entangling gates offer an alternative to conventional sideband-resolved interactions by using ultrafast laser pulses to excite multiple collective motional modes [28]. By designing computer-optimised sequences of pulses, entangling operations can, in principle, be performed on timescales shorter than the trapping period, overcoming conventional spectral-resolution constraints and are therefore non-adiabatic. However, the experimental implementation of these fast gates remains challenging due to demanding laser and control requirements.

Early fast gate proposals focused primarily on entangling neighbouring ions, but recent theoretical developments have extended these operations to non-neighbouring pairs in long-ion chains. In particular, Savill-Brown *et al.* [29] demonstrated that high-fidelity

[1] In this case, it is said that the system exhibits all-to-all connectivity meaning that any arbitrary pair of ions can be entangled

entangling gates could be achieved between arbitrary ion pairs in chains of up to 40 ions, establishing all-to-all connectivity for fast gates. Separately, parallel entangling operations have been demonstrated experimentally in trapped-ion systems using adiabatic laser-driven interactions [1]. These developments offer complementary approaches to reducing circuit execution time in that faster gates reduce t_g, while parallelisation can reduce D. However, whether the impulsive fast-gate framework can admit parallel gate operation remains an important, open question that is the focus of this thesis.

A fundamental challenge arises from the collective motional modes shared by the ions, which can mediate crosstalk and severely degrade the performance of the gate. Consequently, designing parallel fast gates is a complicated control problem where crosstalk must be suppressed while still preserving the desired entangling phases between addressed ions. Whether this can be achieved without compromising gate fidelity, duration or control requirements remains an important question, motivating the investigation presented in this thesis.

1.3 Thesis Outline

The remainder of this thesis is organised as follows. Chapter 2 introduces the theoretical background of trapped-ion quantum computing, focusing on the principles of fast entangling gates, their extension to long-ion chains and existing approaches to gate parallelisation. Chapter 3 extends the theory of two-qubit fast gates to simultaneous operations between multiple ion pairs, establishing the framework for the investigations that follow. Chapter 4 investigates whether independently optimised two-qubit fast gates can be naively applied in parallel, examining the resulting crosstalk and identifying the regimes in which this approach is effective. Chapter 5 instead investigates bespoke, numerically optimised parallel gate sequences designed to suppress crosstalk, comparing their fidelity and resource requirements with sequential execution and the naive parallel gates from Chapter 4. Finally, Chapter 6 critically evaluates the findings of this thesis, discusses their implications for scalable trapped-ion quantum computing and identifies directions for future research.

Chapter 2

Background: Trapped-ion quantum computers

In this chapter I detail the necessary and sufficient concepts that will equip a general physics audience to understand and appreciate the results for parallel fast entangling gates presented in subsequent chapters. In § 2.1 I introduce the fundamentals of quantum information processing that will be essential for understanding the gates I will study. Trapped-ion quantum computers are discussed in § 2.2 with a focus on linear Paul traps and the implementation of ions as qubits. I introduce the collective motional dynamics of the ion chain, § 2.3, and the interaction between ions and light, § 2.4 to motivate the design of adiabatic entangling gates in § 2.5. The key focus of this thesis, fast entangling gates, are then motivated by the issues of adiabatic entangling gates in § 2.6. Recent developments in the study of fast gates are summarised in § 2.7 with a review of how current approaches to parallelisation of gates given in § 2.8.

2.1 Quantum information processing

This section introduces basic elements of quantum information processing, quantum bits (qubits) and gates. I follow Nielsen and Chuang, and consequently direct interested readers to their textbook for a full treatment [10].

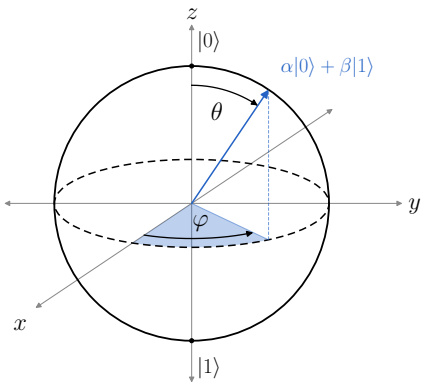


Fig. 2.1: Bloch-sphere representation of a pure qubit state, specified by the polar angle θ and azimuthal angle φ. In this convention, the computational basis state $|0\rangle$ is located at the north pole whilst $|1\rangle$ is at the south pole. A superposition of these two states is understood as any point between the poles.

2.1.1 Qubits

Single qubits

The fundamental unit of classical information is the bit, which can take one of two values, 0 or 1. Its quantum analogue, the qubit, can instead exist in a superposition of the two computational basis states $\{|0\rangle, |1\rangle\}$. Mathematically, a qubit state, $|\psi\rangle$, can be expressed as

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \tag{2.1}$$

where normalisation of the qubit, $\langle\psi|\psi\rangle = 1$, enforces that the complex amplitudes α, β satisfy $|\alpha|^2 + |\beta|^2 = 1$. Geometrically, a pure qubit state can be represented as a point on the unit 2-sphere, known as the Bloch sphere, where $|0\rangle$ represents the north pole and $|1\rangle$ the south pole. This representation motivates an equivalent formulation of Eq. (2.1) in terms of the polar (θ) and azimuthal (φ) angles shown in Fig. 2.1,

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\varphi}\sin\left(\frac{\theta}{2}\right)|1\rangle, \tag{2.2}$$

where an overall global phase has been omitted. This representation is particularly useful when describing the state of a single qubit but is insufficient to describe multi-qubit states, which are essential for the analysis of entangling gates.

Multiple qubits

To understand the space that a *multi-qubit state* lives in, we construct the tensor product space,

$$\mathcal{H}_N = (\mathbb{C}^2)^{\otimes N}, \qquad \dim \mathcal{H}_N = 2^N, \tag{2.3}$$

where $\mathbb{C}^2$ is the Hilbert space of a single qubit. Unlike a classical register of N bits, which occupies one of 2^N possible configurations, an N-qubit system can exist in a superposition of all 2^N computational basis states. It is precisely this exponential scaling of the Hilbert-space dimension that provides quantum computers with a natural framework for efficiently simulating many-body quantum systems [30–32].

Of particular interest in this thesis is the two-qubit state, which can be expressed as a general superposition of the four computational basis states,

$$|\psi\rangle = \alpha'|00\rangle + \beta'|01\rangle + \gamma'|10\rangle + \delta'|11\rangle, \tag{2.4}$$

where we must impose the normalisation condition on the complex amplitudes, $|\alpha'|^2 + |\beta'|^2 + |\gamma'|^2 + |\delta'|^2 = 1$. Here, I have adopted the standard convention for simplifying tensor product notation, $|01\rangle := |0\rangle_1 \otimes |1\rangle_2$, which describes the first qubit in state $|0\rangle$ and the second qubit in state $|1\rangle$.

Entanglement

By studying two-qubit states of the form given in Eq. (2.4), the possibility of encountering a state that cannot be written as the tensor product of two single-qubit states, $|\xi\rangle$ and $|\zeta\rangle$, is introduced. Mathematically, this is the statement that

$$|\psi\rangle \neq |\xi\rangle \otimes |\zeta\rangle, \tag{2.5}$$

for any choice of $|\xi\rangle$ and $|\zeta\rangle$, and such a state is referred to as *entangled*. In contrast, if a two-qubit state can be written as the tensor product of two single-qubit states, it is called *separable*. A fundamental example of an entangled state is the maximally entangled Bell

state,

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}\left(|00\rangle + |11\rangle\right), \tag{2.6}$$

which cannot be expressed as a product of two single-qubit states. Entanglement is a uniquely quantum property that allows correlations between qubits that cannot be described by independent single-qubit states. It is therefore a fundamental resource for quantum information processing and motivates the development of high-fidelity entangling gates [33].

2.1.2 Gates

Quantum computers perform unitary logic operations by controlling qubits. Such operations on qubits will henceforth be referred to as gates, and their performance will be central to the discussion of this thesis. An arbitrary quantum algorithm can be decomposed into two types of gates: single-qubit gates and two-qubit entangling gates [5]. The former are well understood and can be implemented in current quantum computers with errors well below representative fault-tolerance thresholds [34, 35]. It is therefore the implementation of two-qubit entangling gates that remains a significant challenge for current quantum computers. This is to be expected because, unlike single-qubit gates, which correspond to rotations of the qubit state on the Bloch sphere (see Fig. 2.1), an entangling gate requires an interaction between two qubits. Here, I will formally introduce the action of these two types of gates on qubit states and discuss how two-qubit operations can be applied in parallel.

Single-qubit gates

As single-qubit gates act on the two-level qubit state in Eq. (2.1), their action is described by a 2×2 unitary matrix. An important set of such operators is given by the Pauli matrices,

$$\hat{\sigma}_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \hat{\sigma}_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \hat{\sigma}_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{2.7}$$

These operators describe fundamental single-qubit operations and are the generators of rotations about the corresponding axes of the Bloch sphere. For example, applying the

Pauli-X gate to a qubit in state $|0\rangle$ flips it to $|1\rangle$, and vice versa, making it the quantum analogue of the classical NOT gate.

Entangling gates

As only single-qubit gates and two-qubit entangling gates are required for universal quantum computation [5], I will focus on the latter rather than considering general multi-qubit operations. Although gates acting on three or more qubits can also be useful, particularly in quantum error correction, they are not required for the theoretical framework developed here.

A two-qubit gate is described by a 4×4 unitary matrix. Unlike single-qubit operations, entangling gates can transform initially separable states into entangled states. Importantly, entanglement need not arise from a conditional qubit flip; it can also be generated through state-dependent phase accumulation. The Pauli-Z operator leaves the computational basis states unchanged while introducing a relative sign, $\hat{\sigma}_z|0\rangle = |0\rangle$ and $\hat{\sigma}_z|1\rangle = -|1\rangle$. It is therefore convenient to describe state-dependent phase entangling gates in terms of a two-qubit controlled-phase operation generated by $\hat{\sigma}_z^{(A)}\hat{\sigma}_z^{(B)}$,

$$\hat{U}_{\mathrm{CPH}}(\Theta) = \exp\left\{i\Theta\hat{\sigma}_z^{(A)}\hat{\sigma}_z^{(B)}\right\} = \begin{pmatrix} e^{i\Theta} & 0 & 0 & 0 \\ 0 & e^{-i\Theta} & 0 & 0 \\ 0 & 0 & e^{-i\Theta} & 0 \\ 0 & 0 & 0 & e^{i\Theta} \end{pmatrix}. \tag{2.8}$$

Here, $\hat{\sigma}_z^{(i)}$ denotes the Pauli-Z operator acting on the i-th qubit. To demonstrate that this operation can generate entanglement, consider the initially separable state $|\psi_0\rangle = |+\rangle \otimes |+\rangle$, where $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$. Applying Eq. (2.8) with $\Theta = \pi/4$ gives

$$\hat{U}_{\mathrm{CPH}}\left(\frac{\pi}{4}\right)|\psi_0\rangle = \frac{1}{2}\left[e^{i\pi/4}(|00\rangle + |11\rangle) + e^{-i\pi/4}(|01\rangle + |10\rangle)\right]. \tag{2.9}$$

This state is maximally entangled and is equivalent to a Bell state under single-qubit rotations. Therefore, the choice $\Theta = \pi/4$ produces a maximally entangling operation and will be the two-qubit gate studied throughout this thesis.

Gate fidelity

Implementing a quantum computer requires a physical system in which to encode qubits and external control fields to manipulate their states and mediate interactions that realise quantum gates. In practice, imperfections in these controls mean that the physical gate unitary generally only approximates the ideal gate unitary. The *fidelity* quantifies the quality of this approximation by measuring the overlap between the desired and physically realised quantum states.

Suppose an initial state $|\psi_0\rangle$ is acted upon by a physical unitary U_{ph}, which aims to implement the ideal unitary U_{ideal}. The resulting states are $|\psi_{\text{ph}}\rangle = U_{\text{ph}}|\psi_0\rangle$ and $|\psi_{\text{ideal}}\rangle = U_{\text{ideal}}|\psi_0\rangle$, respectively. The state-dependent fidelity is then defined as the squared overlap between these states,

$$F = |\langle\psi_{\text{ph}}|\psi_{\text{ideal}}\rangle|^2 = |\langle\psi_0|U_{\text{ph}}^\dagger U_{\text{ideal}}|\psi_0\rangle|^2. \tag{2.10}$$

However, this fidelity depends on the initial state $|\psi_0\rangle$ and is therefore not indicative of the gate's action on an arbitrary input state. To resolve this, it is convenient to average over all possible initial pure states using the state-averaged gate fidelity, $\mathcal{F}$ [36],

$$\mathcal{F} = \frac{\int_{\psi_0} |\langle\psi_0|U_{\text{ph}}^\dagger U_{\text{ideal}}|\psi_0\rangle|^2 \mathrm{d}|\psi_0\rangle}{\int_{\psi_0} \mathrm{d}|\psi_0\rangle}. \tag{2.11}$$

where $\int_{\psi_0} \mathrm{d}|\psi_0\rangle$ is a normalisation factor. It should be noted that there are various techniques for computing the state-averaged fidelity due to how one defines the differential, $\mathrm{d}|\psi_0\rangle$[1]. For example, for a single qubit this corresponds to averaging uniformly over the Bloch sphere,

$$\int f(\psi_0)\mathrm{d}|\psi_0\rangle = \int_0^\pi \sin(\theta)\mathrm{d}\theta \int_0^{2\pi} f(\theta,\phi)\mathrm{d}\phi. \tag{2.12}$$

The state-averaged fidelity therefore provides a single measure of gate performance that is independent of the choice of initial state. Computationally, it is also very cheap to compute and will serve as the cost function that the optimisation routine will seek to minimise.

[1]This differential is formally referred to as the measure and is fundamental in the study of Lebesgue integration. Readers who are interested in the mathematical details should consult Ref. [37].

The fidelity required for fault-tolerant quantum computation depends on the error-correcting code, noise model and implementation, with representative threshold error rates of order 10^{-2} [38, 39]. Nevertheless, substantially lower physical error rates are desirable for reducing logical errors and the overhead associated with quantum error correction [40]. For this thesis, I therefore adopt a fidelity exceeding 99.9% as a stringent benchmark for identifying promising gate schemes, rather than as a universal requirement for fault tolerance. It is often more convenient to minimise the infidelity, $1 - \mathcal{F}$, which will be used throughout this thesis as $\mathcal{F}$ is often close to unity for the high-fidelity gates considered.

2.2 Trapped-ion quantum computers

This thesis studied trapped-ion quantum computers, where the qubits are encoded in the electronic states of electromagnetically confined atomic ions. Here, I introduce both the trapping potential and qubit ions.

2.2.1 Linear Paul traps

To successfully perform operations on trapped-ion qubits, strict control over their positions is required. A common technique for trapping ions involves the use of electric fields. However, as the electric potential, Φ, must satisfy Laplace's equation in a charge-free region,

$$\nabla^2\Phi = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right)\Phi = 0, \tag{2.13}$$

static electric fields are insufficient to form a three-dimensional trapping region [41], as this would require positive curvature in all three spatial directions. This is overcome by making use of *Paul traps*, which combine static and oscillating electric fields to produce an effective confining potential. Alternatively, *Penning traps* use static electric and magnetic fields to confine ions. Paul traps are widely used in trapped-ion quantum computing [42] and form the trapping architecture considered throughout this work.

Paul traps can support long ion chains by using static electric fields to provide axial (z) confinement, whilst oscillating radiofrequency (RF) electric fields provide radial (x

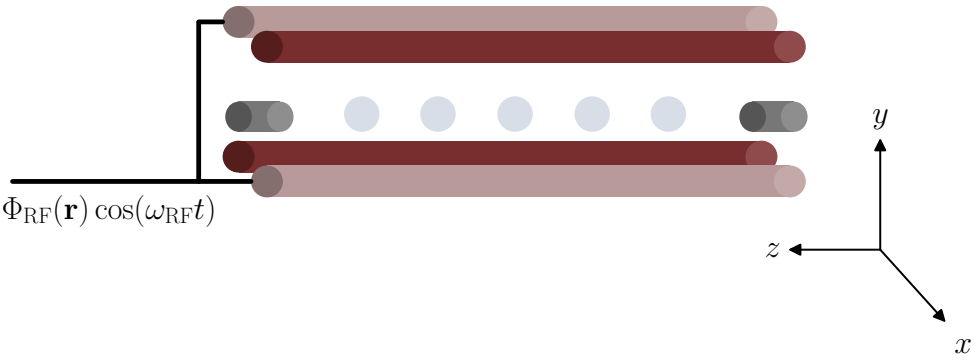


Fig. 2.2: An example of a linear Paul trap. This particular trap is composed of four electrodes (long rods) where the lighter electrodes provide a voltage that oscillates with frequency ω_{RF}. The oscillating RF electric fields provide effective radial confinement in the x–y plane, while static endcap potentials confine the ions along the axial z direction, allowing the formation of a linear ion chain.

and y) confinement. Such implementations are referred to as *linear Paul traps*, with the trapping geometry illustrated in Fig. 2.2. The time-dependent electric potential can be written as [43]

$$\Phi(\mathbf{r},t) = \Phi_{\mathrm{DC}}(\mathbf{r}) + \Phi_{\mathrm{RF}}(\mathbf{r})\cos(\Omega_{\mathrm{RF}}t), \tag{2.14}$$

where Φ_{DC} is the static potential, Φ_{RF} is the spatial amplitude of the oscillating potential, and Ω_{RF} is the angular frequency of the RF drive. The oscillating field produces effective confinement in the radial directions, while the static potential confines the ions axially.

On timescales much longer than the RF period, $2\pi/\Omega_{\mathrm{RF}}$, the slower ion motion can be described using the *pseudopotential approximation*, in which the time-dependent confinement is replaced by an effective static potential [43]. Near the trapping minimum, this potential can be approximated as harmonic,

$$V_{\mathrm{trap}} = \frac{m}{2}\left(\omega_x^2 x^2 + \omega_y^2 y^2 + \omega_z^2 z^2\right), \tag{2.15}$$

where m is the ion mass and ω_x, ω_y and ω_z are the secular trapping frequencies. For a stable linear ion chain, the radial confinement is typically much stronger than the axial confinement, such that $\omega_x, \omega_y \gg \omega_z$. The RF drive also induces rapid radial oscillations, known as *micromotion*, superimposed on the slower secular motion [43]. In an ideal linear Paul trap, ions lie along the RF null, minimising excess micromotion. As the

gates considered throughout this thesis couple exclusively to axial motion, micromotion is neglected.

For multiple ions, the strong radial confinement is essential in preventing their mutual Coulomb repulsion from destabilising the linear configuration. If the confinement is too weak, the chain can undergo a structural transition to a zigzag configuration [44, 45]. For a harmonically confined chain of N ions, an approximate stability criterion is

$$\frac{\omega_r}{\omega_z} \gtrsim 0.63N^{0.865}, \tag{2.16}$$

where $\omega_r = \min(\omega_x, \omega_y)$ is the weaker radial trapping frequency. For fixed radial confinement, increasing the number of ions therefore generally requires weaker axial confinement to maintain a linear chain [46].

2.2.2 Qubit ions

By trapping ions in a linear Paul trap, the goal is to exploit their internal electronic states to encode quantum information. This requires defining a two-level system with qubit states $\{|0\rangle, |1\rangle\}$, separated by a frequency splitting ω_q, corresponding to an energy of $\hbar\omega_q$. The Hamiltonian associated with this system is

$$\hat{H}_q = -\frac{\hbar\omega_q}{2}\hat{\sigma}_z, \tag{2.17}$$

provided that the dynamics can be restricted to the two chosen states. Transitions between these states can be coherently driven through interactions with electromagnetic radiation [43]. This allows the qubit states to be manipulated using external control fields, although unwanted off-resonant couplings and photon scattering can introduce errors that reduce gate fidelity [8]. The interaction between the ion's internal and motional states and an applied laser field is developed later in § 2.4.

Trapped-ion qubits are commonly encoded using either optical transitions between electronic states or hyperfine states within the electronic ground-state manifold [8]. These two encoding schemes are illustrated in Fig. 2.3. Optical qubits permit direct laser-driven transitions between a ground state and a metastable excited state, typically separated

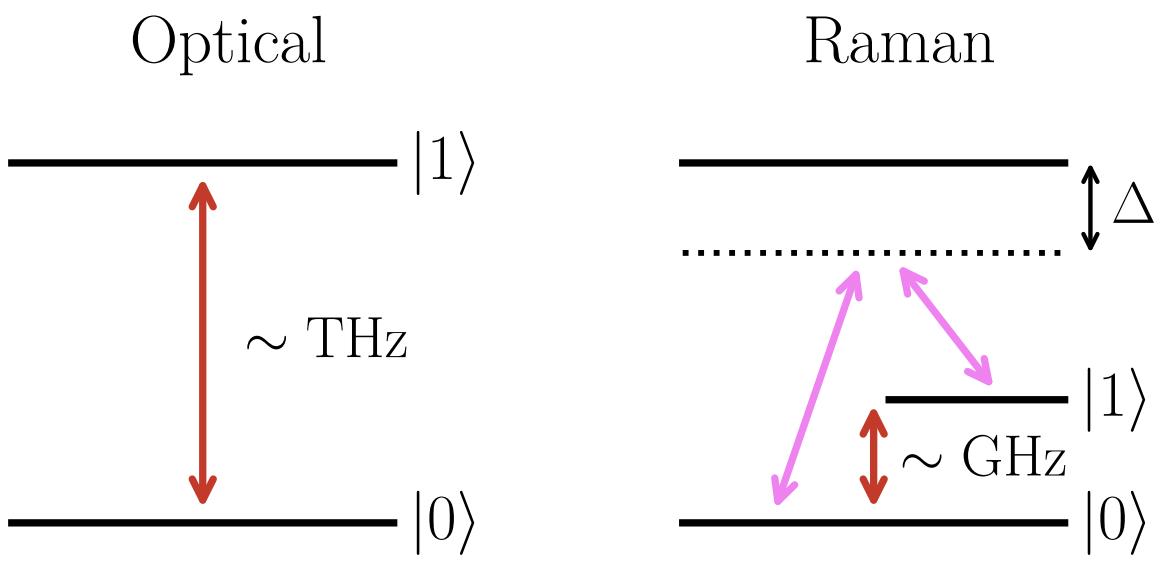


Fig. 2.3: Energy-level diagrams for two common trapped-ion qubit encodings. (a) Optical qubits encode quantum information in a ground state and a metastable excited electronic state, separated by terahertz frequencies, coupled through a direct optical transition. (b) Hyperfine qubits encode information in two levels of the electronic ground-state manifold, which can be coupled through a two-photon stimulated Raman transition. The Raman fields are detuned by Δ from an intermediate excited state, with their frequency difference approximately matching the qubit splitting ω_q and is on the order of gigahertz.

by hundreds of terahertz, but require highly stable optical-frequency control [47]. Hyperfine qubits are another common choice of encoding due to the long coherence times of ground-state hyperfine levels, which are typically separated by frequencies on the order of gigahertz. Transitions between these states can be driven through a two-photon stimulated Raman process, involving the absorption of a photon from one laser field and stimulated emission into another, with both fields detuned from an intermediate excited electronic state [43].

Commonly studied trapped-ion qubits include $^{40}Ca^{+}$ [48], $^{9}Be^{+}$ [49], $^{171}Yb^{+}$ [50] and $^{133}Ba^{+}$ [51, 52]. This thesis considers $^{133}Ba^{+}$, whose nuclear spin $I = 1/2$ provides a simple hyperfine structure with magnetically insensitive clock states and whose visible-wavelength transitions support high-fidelity state preparation and readout [52]. It is also compatible with the 532 nm Raman-driven momentum kicks considered in recent fast-gate proposals, providing an experimentally motivated platform for the schemes studied here [29].

2.3 Collective motional dynamics

The utility of trapping multiple ions in a linear Paul trap comes from their collective motion, which allows entanglement to be mediated between arbitrary ions in the chain [53, 54]. The effective potential experienced by N ions includes both the trapping potential and their mutual Coulomb interactions. Assuming the ions form a linear chain along the axial direction, such that $x_i = y_i = 0$, this potential is given by

$$V_{\text{eff}} = \underbrace{\frac{m\omega_z^2}{2}\sum_{i=1}^{N} z_i^2}_{V_{\text{trap}}} + \underbrace{\frac{e^2}{4\pi\epsilon_0}\sum_{i<j}\frac{1}{|z_i - z_j|}}_{V_{\text{Coulomb}}}. \tag{2.18}$$

Here, z_i is the axial position of the i-th ion, and only the axial trapping frequency, ω_z, is considered due to the strong radial confinement. Physically, the trapping potential confines the ions towards the trap centre, whilst Coulomb repulsion separates them, establishing their equilibrium positions, $z_i^{(0)}$.

The strong confinement of trapped ions means that their motional displacements are typically small compared with their equilibrium separations. For the parameters considered here, ion displacements are on the order of 10 nm, whereas neighbouring ions are separated by approximately 3 µm to 5 µm. Consequently, the dynamics can be well approximated by small oscillations about equilibrium, allowing the nonlinear Coulomb potential to be expanded to second order [55, 56]. Writing the displacement of each ion as $\delta z_i = z_i - z_i^{(0)}$, the potential becomes

$$V_{\text{eff}} \simeq V_{\text{eff}}(\mathbf{z}^{(0)}) + \frac{m}{2}\sum_{i,j=1}^{N} H_{ij}\delta z_i \delta z_j, \qquad H_{ij} = \frac{1}{m}\frac{\partial^2 V_{\text{eff}}}{\partial z_i \partial z_j}\bigg|_{\mathbf{z}=\mathbf{z}^{(0)}}, \tag{2.19}$$

where the linear terms vanish at equilibrium and H_{ij} are the elements of the mass-normalised Hessian matrix with $\mathbf{z}$ as the set of the axial position of the ions. Diagonalising this matrix gives the collective motional modes through the eigenvalue equation [46],

$$\sum_{j=1}^{N} H_{ij} b_\alpha^{(j)} = \omega_\alpha^2 b_\alpha^{(i)}, \qquad \alpha = 1, \ldots, N, \tag{2.20}$$

where ω_α is the frequency of mode α and $b_\alpha^{(i)}$ is the corresponding normalised eigenvector



Fig. 2.4: Axial normal modes of two trapped ions (represented as blue circles). In this case, there are two possible modes in which the ions can oscillate. (a) In-phase motion in the centre-of-mass mode and (b) out-of-phase motion in the breathing mode. Arrows indicate the instantaneous directions of motion.

component, describing how strongly ion i participates in that mode.

By treating the system in this normal-mode basis, an N-ion chain can be described as N independent harmonic oscillators, each oscillating at a frequency ω_α. In the simplest case of $N = 2$, the two axial modes are the centre-of-mass (in-phase) and breathing (out-of-phase) modes, illustrated in Fig. 2.4. The displacement of the i-th ion from equilibrium can therefore be expressed in terms of the normal-mode position operators $\hat{X}_\alpha$,

$$\delta\hat{z}_i = \sum_{\alpha=1}^{N} b_\alpha^{(i)} \hat{X}_\alpha. \tag{2.21}$$

Each normal mode behaves as a quantum harmonic oscillator, with the motional Hamiltonian

$$\hat{H}_{\mathrm{mot},\alpha} = \hbar\omega_\alpha \left(\hat{a}_\alpha^\dagger \hat{a}_\alpha + \frac{1}{2}\right), \tag{2.22}$$

where $\hat{a}_\alpha^\dagger$ and $\hat{a}_\alpha$ are the bosonic creation and annihilation operators, respectively, satisfying $[\hat{a}_\alpha, \hat{a}_\beta^\dagger] = \delta_{\alpha\beta}$ and $[\hat{a}_\alpha, \hat{a}_\beta] = [\hat{a}_\alpha^\dagger, \hat{a}_\beta^\dagger] = 0$. Each excitation corresponds to a phonon of energy $\hbar\omega_\alpha$. Writing the normal-mode position operators in terms of these ladder operators gives

$$\delta\hat{z}_i = \sum_{\alpha=1}^{N} b_\alpha^{(i)} \underbrace{\sqrt{\frac{\hbar}{2m\omega_\alpha}} \left(\hat{a}_\alpha^\dagger + \hat{a}_\alpha\right)}_{\hat{X}_\alpha}. \tag{2.23}$$

The total motional Hamiltonian is then $\hat{H}_{\mathrm{mot}} = \sum_{\alpha=1}^{N} \hat{H}_{\mathrm{mot},\alpha}$, giving the free-evolution unitary over a gate duration t_g,

$$\hat{U}_{\mathrm{mot}} = \prod_{\alpha=1}^{N} \exp\left\{-i\omega_\alpha t_g \hat{a}_\alpha^\dagger \hat{a}_\alpha\right\}. \tag{2.24}$$

Applying a state-dependent force to one ion induces motion in the other ions through the coupled motional modes. In doing this, any two ions in the chain are able to interact. To employ this mechanism for quantum logic operations, the internal qubit states must be coupled to these motional degrees of freedom. I develop the theory of ion–light interactions that enables this coupling in the following section.

2.4 Ion–light interactions

In trapped-ion systems, external electromagnetic fields are used to control the internal qubit states and couple them to the collective motional modes introduced in § 2.3, thereby enabling entangling interactions between ions. For electric-dipole-allowed transitions, this interaction is described by the electric-dipole Hamiltonian [43, 57],

$$\hat{H}_I = -\hat{\mathbf{d}} \cdot \mathbf{E}(\mathbf{r}, t), \tag{2.25}$$

where $\hat{\mathbf{d}} = -e\hat{\mathbf{r}}_e$ is the electric-dipole operator, with $\hat{\mathbf{r}}_e$ denoting the electron's position relative to the nucleus of the ion and $\mathbf{E}(\mathbf{r}, t)$ is the electric field evaluated at the position of the ion. The electric-dipole approximation assumes that the extent of the electronic wavefunction is much smaller than the wavelength of the applied field, allowing the field to be treated as uniform across the electronic charge distribution. Assuming a classical plane-wave field, the electric field is given by

$$\mathbf{E}(\mathbf{r}, t) = \boldsymbol{\epsilon} E_0 e^{i(\mathbf{k}\cdot\mathbf{r} - \omega_L t + \phi_L)} + \text{H.c.}, \tag{2.26}$$

where E_0 is the positive-frequency field amplitude, $\boldsymbol{\epsilon}$ is the unit polarisation vector (taken here to be real), $\mathbf{k}$ is the wavevector, ω_L is the laser angular frequency and ϕ_L is its optical phase.

Restricting the interaction to the two qubit states introduced in § 2.2.2, the coupling strength is characterised by the *Rabi frequency*,

$$\Omega = \frac{2E_0}{\hbar} \left| \langle 1 | \hat{\mathbf{d}} \cdot \boldsymbol{\epsilon} | 0 \rangle \right|. \tag{2.27}$$

Projecting Eq. (2.25) onto the two-level system and absorbing the phase of the transition matrix element into ϕ_L gives

$$\hat{H}_I = \frac{\hbar\Omega}{2}(\hat{\sigma}_+ + \hat{\sigma}_-)\left[e^{i(k\hat{x}-\omega_L t+\phi_L)} + e^{-i(k\hat{x}-\omega_L t+\phi_L)}\right], \tag{2.28}$$

where $\hat{x}$ denotes displacement along the laser wavevector, taken to be aligned with the axial direction of the trap. Here, $\hat{\sigma}_+ = |1\rangle\langle 0|$ and $\hat{\sigma}_- = (\hat{\sigma}_+)^\dagger = |0\rangle\langle 1|$ are the qubit raising and lowering operators.

The interaction Hamiltonian can be transformed into the interaction picture with respect to the qubit Hamiltonian in Eq. (2.17). Defining $\hat{U}_0(t) = \exp(-i\hat{H}_q t/\hbar)$, the transformation is

$$\hat{H}_I^{(I)}(t) = \hat{U}_0^\dagger(t)\hat{H}_I(t)\hat{U}_0(t). \tag{2.29}$$

Using $\hat{U}_0^\dagger\hat{\sigma}_\pm\hat{U}_0 = e^{\pm i\omega_q t}\hat{\sigma}_\pm$, the transformed Hamiltonian becomes

$$\begin{aligned}\hat{H}_I = \frac{\hbar\Omega}{2}\Big[&\hat{\sigma}_+ e^{i(k\hat{x}-\Delta t+\phi_L)} + \hat{\sigma}_- e^{-i(k\hat{x}-\Delta t+\phi_L)} \\ &+ \hat{\sigma}_+ e^{-i[k\hat{x}-(\omega_L+\omega_q)t+\phi_L]} + \hat{\sigma}_- e^{i[k\hat{x}-(\omega_L+\omega_q)t+\phi_L]}\Big],\end{aligned} \tag{2.30}$$

where $\Delta = \omega_L - \omega_q$ is the detuning of the driving field from the qubit transition. For near-resonant driving, the terms oscillating at $\omega_L+\omega_q$ can be neglected under the *rotating-wave approximation* (RWA), as these oscillations are rapid ($\sim$THz) compared with the relevant interaction frequencies ($\sim$MHz). For direct optical transitions, these frequencies are typically on the order of hundreds of terahertz, far exceeding the megahertz-scale secular motion and conventional Rabi frequencies. Neglecting these rapidly oscillating terms gives

$$\hat{H}_\mathrm{I}' = \frac{\hbar\Omega}{2}\left(\hat{\sigma}_+ e^{i(k\hat{x}-\Delta t+\phi_L)} + \hat{\sigma}_- e^{-i(k\hat{x}-\Delta t+\phi_L)}\right). \tag{2.31}$$

The validity of the RWA for impulsive fast-gate protocols has been investigated by Bentley et al. [58], who demonstrated its robustness with respect to the number of applied pulses.

Importantly, the spatial dependence $e^{ik\hat{x}}$ in Eq. (2.31) couples the internal qubit states to the ion's motion. In particular, the operator $e^{ik\hat{x}}$ translates a motional momentum eigenstate according to $e^{ik\hat{x}}|p\rangle = |p+\hbar k\rangle$, such that the qubit transition induced by

$\hat{\sigma}_+$ is accompanied by a momentum transfer of $+\hbar k$. This coupling between internal-state transitions and motional momentum forms the basis of the state-dependent kicks considered later in this thesis.

To quantify the strength of the spin–motion coupling, I introduce the *Lamb–Dicke parameter* for the axial centre-of-mass mode of frequency ω_z,

$$\eta = k\sqrt{\frac{\hbar}{2m\omega_z}} \implies k\hat{x} = \eta(\hat{a} + \hat{a}^\dagger), \tag{2.32}$$

which allows the interaction Hamiltonian to be expressed as

$$\hat{H}_\mathrm{I}' = \frac{\hbar\Omega}{2}\left(\hat{\sigma}_+ e^{i(\phi_L - \Delta t)} e^{i\eta(\hat{a}+\hat{a}^\dagger)} + \hat{\sigma}_- e^{-i(\phi_L-\Delta t)} e^{-i\eta(\hat{a}+\hat{a}^\dagger)}\right). \tag{2.33}$$

For a chain of ions, this generalises to the collective motional modes through the normal-mode expansion in Eq. (2.23). The Lamb–Dicke approximation corresponds to expanding the position-dependent exponentials when the motional excursions are sufficiently small. In contrast, impulsive fast-gate protocols retain the full exponential dependence, allowing operation outside this approximation. These distinct regimes are examined in the following sections.

2.5 Adiabatic entangling gates

Conventional sideband-resolved trapped-ion entangling gates, here referred to as *adiabatic gates*, operate in the weak-driving regime, allowing selective excitation of motional sidebands [24, 25, 59]. These gates are typically described within the Lamb–Dicke regime, which requires the ion's motional excursions to be much smaller than the wavelength of the driving field. For a single motional mode, the Lamb–Dicke criterion is [43, 57]

$$\eta\sqrt{\langle(\hat{a} + \hat{a}^\dagger)^2\rangle} \ll 1, \tag{2.34}$$

which also implies the weaker condition $\eta \ll 1$. When this criterion is satisfied, the spatial exponentials in Eq. (2.33) can be expanded to first order. Generalising to an N-ion chain

using Eq. (2.23), the position of the i-th ion along the driving direction is

$$k\hat{x}_i = \sum_{\alpha=1}^{N} \eta_\alpha b_\alpha^{(i)} (\hat{a}_\alpha + \hat{a}_\alpha^\dagger), \tag{2.35}$$

where $\eta_\alpha = k\sqrt{\hbar/(2m\omega_\alpha)}$ is the mode-dependent Lamb–Dicke parameter, characterising the coupling strength to mode α. Transforming into the interaction picture with respect to the motional Hamiltonian, Eq. (2.22), gives the first-order interaction Hamiltonian [43],

$$\hat{H}_I' \simeq \frac{\hbar\Omega}{2} \hat{\sigma}_+^{(i)} e^{i(\phi_L - \Delta t)} \left[1 + i \sum_{\alpha=1}^{N} \eta_\alpha b_\alpha^{(i)} \left(\hat{a}_\alpha e^{-i\omega_\alpha t} + \hat{a}_\alpha^\dagger e^{i\omega_\alpha t} \right) \right] + \text{H.c.}, \tag{2.36}$$

where the first term describes the carrier transition, while $\hat{a}_\alpha$ and $\hat{a}_\alpha^\dagger$ describe the red and blue motional sidebands, respectively, which become resonant for detunings $\Delta = -\omega_\alpha$ and $\Delta = +\omega_\alpha$.

Selective excitation of these sidebands requires sufficiently weak driving, $\Omega \ll \omega_z$, to suppress unwanted carrier excitation, with further restrictions arising from closely spaced motional modes. Equivalently, a laser pulse of duration T has a spectral bandwidth of order $1/T$, which must be smaller than the frequency separation between neighbouring sidebands. Thus, spectral selectivity requires relatively long interaction times, constraining the achievable gate speed [60, 61]. A prominent example is the Mølmer–Sørensen gate, which uses bichromatic fields near the red and blue sidebands to generate a spin-dependent force [24, 25]. This force drives spin-dependent trajectories through motional phase space, accumulating geometric phases while restoring the motion at the end of the gate.

Fast gates instead operate in the strong, impulsive driving regime, using ultrafast laser pulses to impart state-dependent momentum kicks on timescales shorter than the characteristic motional periods [28]. These protocols do not require the Lamb–Dicke approximation and retain the full spatial dependence $e^{ik\hat{x}_i}$ of the ion–light interaction. As follows from the normal-mode expansion above, each momentum kick couples to multiple motional modes, with the short pulse duration producing broadband rather than spectrally selective excitation. The strengths and timings of these kicks are then chosen to restore the coupled motional modes whilst accumulating the desired entangling phase.

The following section develops this approach.

2.6 Fast entangling gates

Since their original proposal by García-Ripoll *et al.* [28], fast entangling gates have been extensively investigated theoretically [60, 62–64]. Unlike adiabatic sideband-resolved gates, such as the Mølmer and Sørensen gate, fast gates exploit multiple motional modes and can operate on timescales shorter than the trap period, overcoming the conventional adiabatic restriction. Experimental demonstrations include entanglement generation using ultrafast optical pulses [50] and high-fidelity non-adiabatic gates using amplitude-shaped pulses [11][2]. However, achieving high fidelities with impulsive ultrafast-pulse protocols remains experimentally challenging [50].

The impulsive fast-gate protocols considered in this thesis use sequences of ultrafast laser pulses to impart spin-state-dependent momentum kicks (SDKs) to trapped ions. These kicks excite collective motional modes, generating spin-state-dependent trajectories whose accumulated geometric phases produce entanglement. In principle, the gate speed is not fundamentally restricted by the trap period, although practical implementations are constrained by available laser power, pulse repetition rates and control precision [28, 65].

2.6.1 State-dependent kicks

Unlike adiabatic sideband-resolved gates, impulsive fast gates do not require the Lamb–Dicke approximation. We therefore retain the full spatial dependence of the ion–laser interaction, without expanding the exponential terms in Eq. (2.31). The ultrafast pulses that are used are typically on, or near, resonance with the internal state of the ion and as such in the following, I set $\Delta = 0$ in Eq. (2.31) to obtain,

$$\hat{H}_{\mathrm{I}}' = \frac{\hbar\Omega}{2}\left(\hat{\sigma}_{+}e^{i(k\hat{x}+\phi_L)} + \hat{\sigma}_{-}e^{-i(k\hat{x}+\phi_L)}\right). \tag{2.37}$$

[2]Although operating in the non-adiabatic regime, the protocol of Schäfer *et al.* [11] differs from the impulsive SDK-based approach considered in this thesis, as it employs shaped laser driving rather than discrete momentum kicks.

An ultrafast resonant π-pulse, satisfying $\Omega t = \pi$ for constant Ω, inverts the internal state while imparting momentum $\pm\hbar k$, depending on the initial qubit state. As SDKs are delivered on timescales much shorter than the trap period, motional evolution during the pulse is negligible, and hence, the corresponding unitary is, up to a global phase,

$$\hat{U}_\pi = \begin{pmatrix} 0 & e^{-i(k\hat{x}+\phi_L)} \\ e^{i(k\hat{x}+\phi_L)} & 0 \end{pmatrix} = \hat{\sigma}_x e^{i(k\hat{x}+\phi_L)\hat{\sigma}_z}. \tag{2.38}$$

In practice, an SDK is constructed from two counter-propagating π-pulses applied in rapid succession. The second pulse restores the internal state while imparting an additional momentum kick in the same direction as the first. Assuming equal optical phases and negligible motional evolution between pulses, their combined unitary is

$$\hat{U}_{\pi\text{-pair}} = \hat{U}_{-k}\hat{U}_k = \exp\left\{2ik\hat{x}\hat{\sigma}_z\right\}, \tag{2.39}$$

which is independent of the common laser phase ϕ_L. The resulting momentum transfer of $\pm 2\hbar k$, determined by the initial qubit state, is illustrated in Fig. 2.5.

To construct a two-qubit fast gate, SDKs are applied to ions A and B, which are coupled through their collective motion. For a pulse group containing $|z|$ identical SDKs, the combined unitary is

$$\hat{U}_{\text{SDK}} = \exp\left\{2izk\left(\hat{x}_A\hat{\sigma}_z^{(A)} + \hat{x}_B\hat{\sigma}_z^{(B)}\right)\right\}, \tag{2.40}$$

where the sign of z determines the direction of the state-dependent momentum transfer. Expressing the ion displacements in terms of the normal modes introduced in § 2.3 which take the form given in Eq. (2.23), and using the displacement operator $\hat{D}_\alpha(\beta) = \exp\{\beta\hat{a}_\alpha^\dagger - \beta^*\hat{a}_\alpha\}$, gives

$$\begin{aligned}\hat{U}_{\text{SDK}} &= \prod_{\alpha=1}^{N} \exp\left\{2iz\eta_\alpha\left(b_\alpha^{(A)}\hat{\sigma}_z^{(A)} + b_\alpha^{(B)}\hat{\sigma}_z^{(B)}\right)\left(\hat{a}_\alpha + \hat{a}_\alpha^\dagger\right)\right\} \\ &= \prod_{\alpha=1}^{N} \hat{D}_\alpha\left(2iz\eta_\alpha\left(b_\alpha^{(A)}\hat{\sigma}_z^{(A)} + b_\alpha^{(B)}\hat{\sigma}_z^{(B)}\right)\right),\end{aligned} \tag{2.41}$$

where η_α is the mode-dependent Lamb–Dicke parameter.

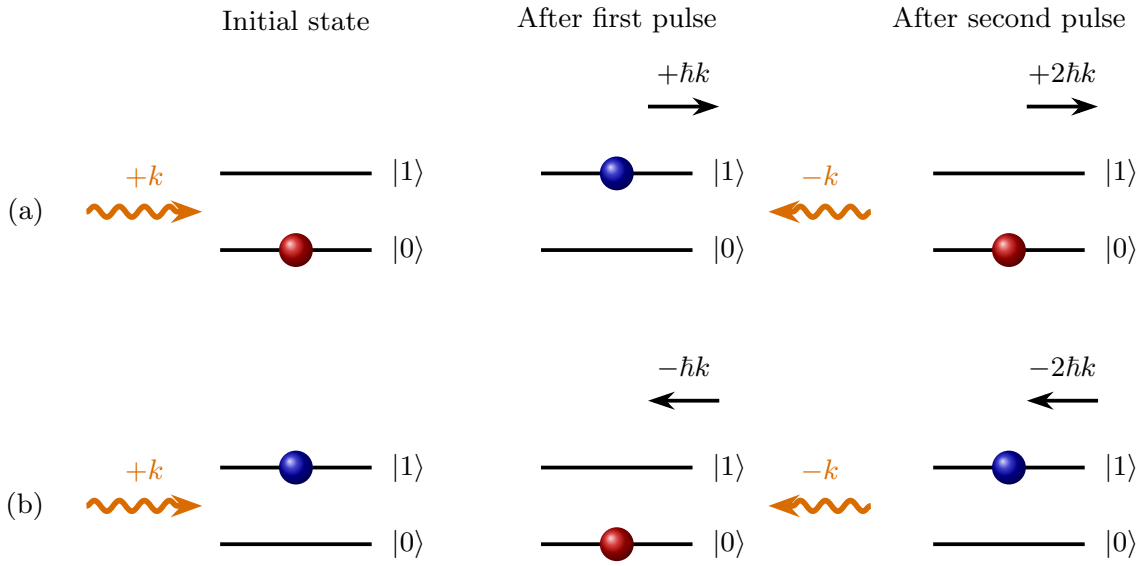


Fig. 2.5: State-dependent momentum transfer induced by two counter-propagating resonant π-pulses with wavevectors $+k$ and $-k$. (a) An ion initially in $|0\rangle$ undergoes $|0\rangle \to |1\rangle \to |0\rangle$, acquiring a net momentum of $+2\hbar k$. (b) An ion initially in $|1\rangle$ undergoes the reverse transitions, acquiring $-2\hbar k$. In both cases, the internal state is restored while the momentum transfer depends on the initial qubit state, forming the state-dependent kicks that are essential for the implementation of the impulsive fast gates studied here.

Although an individual SDK does not generate entanglement, successive SDKs separated by free motional evolution produce spin-dependent trajectories through phase space. The non-commutativity of these displacements generates a geometric phase, allowing the ions to become entangled even when the net motional displacement vanishes. These two requirements, entangling-phase accumulation and motional restoration, form the basis of the fast-gate theory developed in the following section.

2.6.2 Fast gate theory

Fast gates use SDKs which are realised as pairs of counter-propagating π-pulses that drive state-dependent motion of two target ion qubits. A fast gate delivers SDKs in groups, termed pulse groups, separated by periods of free evolution. This evolution rotates the spin-dependent displacements in phase space, such that successive SDKs generally do not commute. The resulting geometric phase contains a spin–spin interaction proportional to $\hat{\sigma}_z^{(A)}\hat{\sigma}_z^{(B)}$, enabling entanglement even when the net motional displacement vanishes [46,

65]. This will be derived explicitly in Chapter 3 for the case of multi-qubit entanglement. The ideal controlled-phase operation between ions A and B is therefore

$$\hat{U}_{\mathrm{CPH}} = \exp\left\{ i\frac{\pi}{4}\hat{\sigma}_z^{(A)} \otimes \hat{\sigma}_z^{(B)} \right\}. \tag{2.42}$$

For a fast gate of duration t_g, combining the desired entangling operation with the free motional evolution in Eq. (2.24) gives the ideal gate unitary,

$$\hat{U}_{\mathrm{ideal}} = \hat{U}_{\mathrm{CPH}}\hat{U}_{\mathrm{mot}} = \exp\left\{ i\frac{\pi}{4}\hat{\sigma}_z^{(A)} \otimes \hat{\sigma}_z^{(B)} \right\} \prod_{\alpha=1}^{N} \exp\left\{ -i\omega_\alpha t_g \hat{a}_\alpha^\dagger \hat{a}_\alpha \right\}. \tag{2.43}$$

An ideal fast gate must therefore accumulate the desired entangling phase while restoring the motional modes relative to their free evolution, leaving no residual spin–motion entanglement.

2.6.3 Fast gate condition equations

As noted in § 2.6.2, implementing the ideal unitary in Eq. (2.43) requires two conditions: accumulating a maximally entangling phase between the target ions whilst simultaneously restoring the motional modes relative to their free evolution. Here, I formalise these conditions mathematically for two-qubit entanglement. An N-ion chain admits N motional modes along the driven direction, and restoring these modes requires N complex condition equations [46, 65],

$$0 = \Lambda_\alpha = 2i\eta_\alpha \sum_{j=1}^{K} z_j e^{i\omega_\alpha t_j}, \tag{2.44}$$

with $\alpha = 1, \ldots, N$. In contrast, the maximally entangling phase must only be accumulated between the internal states of the two target ions, giving a single additional condition equation,

$$\frac{\pi}{4} = \Theta = 8\sum_{\alpha=1}^{N} \eta_\alpha^2 b_\alpha^{(A)} b_\alpha^{(B)} \sum_{m=2}^{K}\sum_{j=1}^{m-1} z_m z_j \sin(\omega_\alpha(t_m - t_j)). \tag{2.45}$$

Finding a sequence of K pulse groups, z_k, delivered at times t_k defines a fast-gate scheme $\{z_k, t_k\}$. By numerically optimising this sequence, one can find solutions that approximately satisfy the $N+1$ condition equations, Eqs. (2.45) and (2.44), corresponding to $2N+1$ real constraints. Assuming small errors and initially thermal motional states, the state-averaged infidelity of an entangling gate acting on ions A and B is given by [46, 65]

$$1 - F \simeq \underbrace{\frac{2}{3}\left|\Theta - \frac{\pi}{4}\right|^2}_{\text{Phase error}} + \underbrace{\frac{4}{3}\sum_{\alpha=1}^{N}\left(\frac{1}{2}+\bar{n}_\alpha\right)\left[\left(b_\alpha^{(A)}\right)^2 + \left(b_\alpha^{(B)}\right)^2\right]|\Lambda_\alpha|^2}_{\text{Motional restoration error}}. \tag{2.46}$$

Here, $\bar{n}_\alpha$ is the initial mean phonon occupation of mode α. The small-error approximation assumes that higher-order contributions in the phase error, $\Theta - \pi/4$, and residual motional displacements, Λ_α, can be neglected. This approximation is routinely employed in fast-gate studies [29, 66, 67], where low-infidelity solutions are sought and higher-order contributions are therefore small. The resulting infidelity provides the cost function for the numerical optimisation of fast-gate sequences discussed in the following section.

2.6.4 Fast-gate optimisation

Constructing high-fidelity fast gates requires finding sequences of state-dependent kicks that accumulate the desired entangling phase while restoring all coupled motional modes. Satisfying these conditions becomes increasingly difficult for longer ion chains, particularly under constraints on gate duration. Consequently, pulse sequences are obtained through numerical optimisation [46, 65]. We consider K pulse groups, each containing $|z_k|$ SDKs arriving at time t_k, where the sign of z_k determines the kick direction. The sequence is parametrised by

$$\mathbf{z} = \{z_1, \ldots, z_K\}, \tag{2.47}$$

$$\mathbf{t} = \{t_1, \ldots, t_K\}, \tag{2.48}$$

giving $2K$ parameters to optimise by minimising the gate infidelity in Eq. (2.46).

The optimisation proceeds in two stages [46, 65]. First, the pulse-group strengths $\mathbf{z}$ are optimised with fixed timings $\mathbf{t}$. Since each z_k represents an integer number of

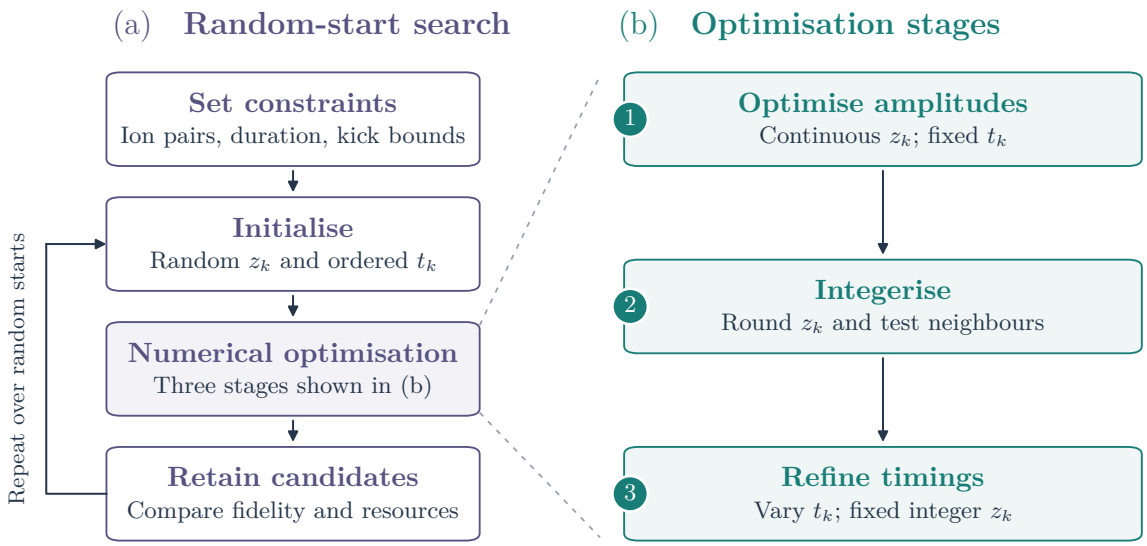


Fig. 2.6: Optimisation routine for designing fast gate schemes. (a) Repeated local searches from random initial conditions retain candidates according to fidelity and SDK resource requirements. (b) Each search optimises according to the two-stage approach. Optimisation of continuous pulse-group strength at fixed timings, are then rounded to integers and neighbouring values are tested as an intermediate stage. Finally, timings are refined at fixed integer SDK strengths.

SDKs, this formally constitutes a nonlinear integer-programming problem. To reduce computational complexity, the integer constraint is relaxed, allowing the strengths to be optimised continuously. The resulting strengths are rounded to integers, after which the timings are optimised with $\mathbf{z}$ fixed. This ordering also exploits the polynomial dependence of the approximate infidelity on z_k, compared with its oscillatory dependence on t_k.

To further reduce the search space, we adopt the antisymmetric pulse group (APG) scheme introduced by Gale *et al.* [65] and subsequently applied to long ion chains by Mehdi *et al.* [46]. Antisymmetry automatically satisfies one quadrature of the motional restoration condition, Eq. (2.44), for every mode. For even K, the scheme takes the form

$$\mathbf{z} = \{-z_{K/2}, \ldots, -z_1, z_1, \ldots, z_{K/2}\}, \tag{2.49}$$

$$\mathbf{t} = \{-t_{K/2}, \ldots, -t_1, t_1, \ldots, t_{K/2}\}, \tag{2.50}$$

where timings are defined relative to the gate midpoint. This reduces the number of independent optimisation parameters from $2K$ to K, simplifying the search for high-

fidelity solutions.

Despite these simplifications, the optimisation landscape remains non-convex, with numerous local minima. Therefore, up to $\sim 10^3$ local searches are performed from different initial conditions, retaining the lowest-infidelity solution found. Although this random initial guess approach improves exploration of the search space, it does not guarantee convergence to the global minimum which is a limitation highlighted in Chapter 5. This optimisation framework provides the basis for the parallel fast-gate searches developed in subsequent chapters, where multiple target interactions must be generated while suppressing crosstalk.

2.7 Fast gates in long ion chains

Early fast-gate proposals established how sequences of state-dependent kicks could entangle a pair of nearest-neighbour trapped ions on timescales shorter than the trap period [28] and hence operate in a regime where the adiabatic condition need not be satisfied. As discussed in the preceding sections, these sequences must accumulate the desired entangling phase while restoring the coupled motional modes. Moreover, although fast-gate protocols excite multiple collective motional modes, the short interaction timescale results in largely localised ion motion. Consequently, original fast-gate schemes favour nearest-neighbour interactions, where the stronger Coulomb coupling enables rapid entanglement generation [60]. Extending this framework to longer chains enables operations on a larger array of qubits within a single trap, reducing the need to transport ions between trapping regions [46, 60]. However, an N-ion chain has N axial motional modes, so increasing the chain length introduces additional restoration conditions and can complicate the optimisation problem outlined in § 2.6.4. Recalling the discussion in § 2.3, maintaining a linear configuration also requires suitable axial confinement, with the onset of buckling depending on both the ion number and the trapping potential [44].

Early fast-gate studies focused primarily on nearest-neighbour ion pairs, where sufficiently short gate durations produce spatially localised motion, allowing favourable scaling with chain length [60]. Mehdi *et al.* [46] subsequently demonstrated high-fidelity gate solutions with durations of approximately 1 µs, without increasing pulse-control require-

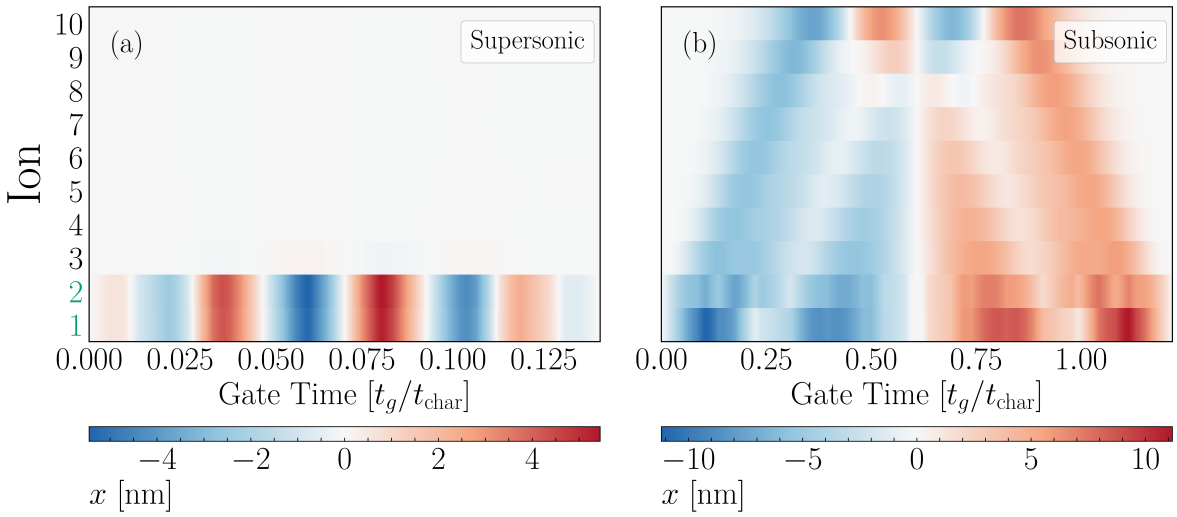


Fig. 2.7: Motional dynamics due to entangling two nearest neighbour ions (ions 1 and 2) in a 10-ion chain. (a) Supersonic and (b) subsonic regimes, with gate durations $t_g = 0.466\,\mu\text{s} < t_{\text{char}}$ and $t_g = 4.041\,\mu\text{s} > t_{\text{char}}$, respectively. The colour indicates the displacement of each ion from its equilibrium position, with red and blue representing positive and negative displacements (note the different scales). The supersonic regime displays highly localised dynamics, with only the addressed ions being displaced due to the SDK sequence. The subsonic regime demonstrates that there is sufficient time for phonons to make a round trip across the chain resulting in non-localised motional dynamics. In both (a) and (b) the gate is normalised by the characteristic timescale t_{char}.

ments for longer chains. Although Bentley *et al.* [60] also investigated non-local gates between ions separated by one or more intervening ions, these required substantially greater momentum transfers and more demanding laser repetition rates.

More recently, Savill-Brown *et al.* [29] demonstrated that collective phonon propagation enables fast entangling gates between arbitrary ion pairs. They characterised this behaviour using the approximate round-trip propagation time of a phonon wavepacket,

$$t_{\text{char}} \simeq \frac{2L}{c_s} \simeq \frac{2\pi}{\Delta\omega_{\text{max}}}, \qquad \Delta\omega_{\text{max}} = \max_{\alpha} |\omega_{\alpha+1} - \omega_{\alpha}|, \tag{2.51}$$

where L is the chain length, c_s is the effective phonon propagation speed, and $\Delta\omega_{\text{max}}$ is the maximum frequency separation between adjacent motional modes. This distinguishes two regimes: *supersonic* gates ($t_g < t_{\text{char}}$), where motion remains largely localised around the driven ions, and *subsonic* gates ($t_g > t_{\text{char}}$), where disturbances propagate throughout the chain. These contrasting dynamics are illustrated in Fig. 2.7.

Exploiting the subsonic regime, Savill-Brown *et al.* [29] identified high-fidelity non-local gate solutions in chains of up to 40 ions, enabling all-to-all connectivity without the demanding pulse requirements of earlier supersonic approaches for non-local gates. In complementary work, Savill-Brown *et al.* [68] further improved experimental feasibility through numerical optimisation, obtaining solutions requiring only tens of pulses under the conditions studied. These developments establish the potential for practical high-connectivity fast gates, although their fidelity remains sensitive to pulse imperfections and motional control.

These results motivate investigating simultaneous operations on multiple pairs. An N-ion chain contains at most $\lfloor N/2 \rfloor$ disjoint pairs, but connectivity alone does not guarantee that their gates can be performed concurrently. Shared motion can generate crosstalk requiring the pulse sequence to satisfy additional entangling phase constraints [1]. Determining whether suitable parallel SDK sequences exist under the available control restrictions is the central focus of this thesis. The following section reviews existing approaches to trapped-ion gate parallelisation and places this question within the broader literature.

2.8 Approaches to parallelisation

Parallel entangling operations offer a route to reducing quantum circuit execution times and the time qubits spend idle [1, 69]. However, all-to-all connectivity does not automatically imply that arbitrary gates can be performed simultaneously. In a trapped-ion chain, the collective motional modes that mediate the desired interactions can also generate crosstalk. Parallelisation therefore requires control of both the target interactions and the associated crosstalk, while ensuring that the spins disentangle from the motion at the end of the operation [1].

One approach to reduce crosstalk between parallel gates involves controlling which motional degrees of freedom mediate each interaction. Zhu *et al.* [69] experimentally demonstrated pairwise parallel gates using orthogonal sets of transverse motional modes. Alternatively, Olsacher *et al.* [70] and Cheng *et al.* [71] proposed using optical tweezers to modify the local confinement and create localised motional modes. These modes allow different subsets of ions to undergo entangling operations in parallel. Such approaches

illustrate how engineering the motional structure can prevent crosstalk but requires increasing the experimental complexity in the form of additional optics and lasers.

To avoid additional laser systems, parallelisation can also be pursued through spatially separated interaction regions. In such architectures, ions are transported between storage and processing regions, allowing operations to be distributed across the device [72, 73]. This provides an alternative to performing all operations within one long chain, although transport, cooling and scheduling must be incorporated into the execution of a gate which make up 95% of the total operating time and therefore imposes significant overheads [27].

Another approach is to segment the driving fields so that multiple gates can share the motional modes while suppressing crosstalk. Figgatt *et al.* [1] experimentally demonstrated parallel two-qubit entangling operations within an ion chain using appropriately designed control pulses. Grzesiak *et al.* [74] subsequently developed efficient arbitrary simultaneously entangling gates, providing a framework for performing multiple pairwise interactions. These results demonstrate that shared motion can support simultaneous operations when the driving fields are designed to satisfy the combined interaction and motional-restoration requirements. Optimising pulse sequences to additionally suppress crosstalk will likely increase the complexity of the search but I nonetheless adopt this approach as the fast-gate optimisation protocol discussed in § 2.6.4 is well suited for this problem. Moreover, alternatives involving spatially separated interaction regions or the excitation of orthogonal sets of motional modes would introduce additional experimental complexity and require extensive re-design of fast gate optimisation process.

These developments motivate investigating parallelisation within the theoretical impulsive fast-gate framework considered in this thesis. The available controls are the strengths z_k and timings t_k of SDKs, which determine both the motional trajectories and the accumulated entangling phases. The central question is whether these impulsive controls can simultaneously restore the coupled motional modes, generate the required entangling phases on multiple target pairs, and suppress crosstalk thereby overcoming the challenges faced in similar control-based approaches for adiabatic gates [1]. The following chapter develops the multi-qubit gate description and fidelity measures needed to formulate this problem, providing the basis for the subsequent numerical search for parallel fast-gate sequences in Chapters 4 and 5.

Chapter 3

Theory of multi-qubit fast gates

The fast-gate theory discussed in Chapter 2 has been highly successful in identifying pulse schemes that generate maximally entangling operations between arbitrary ion pairs, even in long chains [29]. These schemes are typically found through numerical optimisation, requiring an efficient cost function to explore the high-dimensional parameter space of pulse timings, directions and strengths. While existing two-qubit methods aim to restore the shared motional modes while accumulating the desired entangling phase, extending these methods to multiple addressed ions requires accounting for additional pairwise interactions. In this chapter, I generalise the two-qubit fidelity formalism to describe such operations, encompassing multi-qubit entangling gates [75] and specifically the focus of this thesis, parallel two-qubit gates [1].

I begin by defining the target unitary operation in § 3.1, before deriving the unitary evolution generated by SDK sequences applied to multiple ions in § 3.2. I then derive the corresponding state-averaged infidelity using a small-error approximation in § 3.3, providing a computationally efficient measure of how closely an SDK sequence implements the target operation. Although the formalism accommodates an arbitrary number of addressed ions, its application in this thesis focuses on two simultaneous gates acting on disjoint pairs (see Fig. 3.1). This infidelity is used in the following chapters to assess existing two-qubit solutions under parallel execution and to optimise pulse sequences specifically for parallel operation.

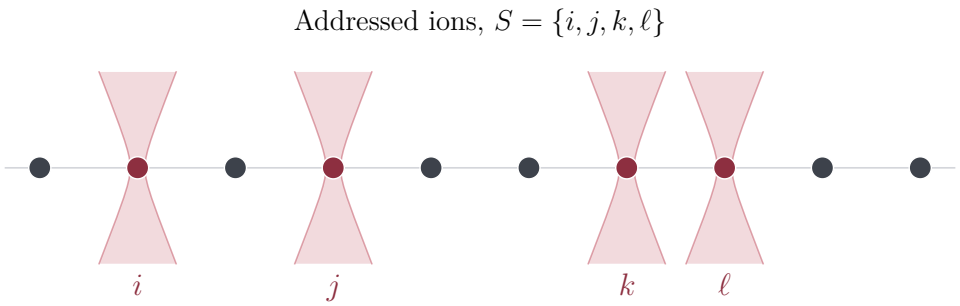


Fig. 3.1: A linear ion chain with four ions addressed by the red cones which represent the laser pulses delivering the SDKs. The set $S = \{i, j, k, \ell\}$ describes the ions that receive SDKs. From the set S, disjoint target pairs for parallel operations can be identified, e.g. (i, j) and (k, ℓ). It should also be noted that not all ions in S must necessarily be addressed simultaneously.

3.1 Target operation for multi-qubit gates

To define the fidelity of a multi-qubit gate operation, we first specify the ideal unitary, $\hat{U}^{\mathrm{MQ}}_{\mathrm{G,ideal}}$, of the unitary gate operation,

$$\hat{U}^{\mathrm{MQ}}_{\mathrm{G,ideal}} = \exp\left[i \sum_{\substack{j,\ell \in S \\ j<\ell}} \Theta^{\mathrm{tar}}_{j\ell} \hat{\sigma}_z^{(j)} \hat{\sigma}_z^{(\ell)} \right]. \tag{3.1}$$

Here, S denotes the set of addressed ions, while $\Theta^{\mathrm{tar}}_{j\ell}$ specifies the target entangling phase between ions j and ℓ. The example of four addressed ions is the set, $S = \{i, j, k, \ell\}$ and is represented schematically in Fig. 3.1. The type of gate is then defined by the target entangling-phase matrix, which can distinguish parallel two-qubit gates from more general multi-qubit entangling operations. The following matrices demonstrate this,

$$\Theta^{\mathrm{tar}} = \frac{\pi}{4} \begin{array}{c} \\ i \\ j \\ k \\ \ell \end{array} \begin{array}{c} \begin{array}{cccc} i & j & k & \ell \end{array} \\ \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \end{array}, \qquad \Theta^{\mathrm{tar}} = \frac{\pi}{4} \begin{array}{c} \\ i \\ j \\ k \\ \ell \end{array} \begin{array}{c} \begin{array}{cccc} i & j & k & \ell \end{array} \\ \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix} \end{array}. \tag{3.2}$$

The first example is for parallel operation as it couples only the disjoint pairs (i, j) and (k, ℓ), whereas the second is instead a $4-$qubit gate as it generates a pairwise entangling operation between every pair of ions. In either case, along with acquiring the target entangling phases, the physical gate must also restore the motional modes of the ions by its completion. As the focus of this thesis is parallel fast gates, I will henceforth consider target entangling-phase matrices that are of the form of the first example, although the theory developed in this chapter can be applied to more general multi-qubit entangling operations shown in the second example.

3.2 Multi-qubit fast-gate evolution

Analogously to two-qubit gates, multi-qubit gates are realised as a sequence of SDKs separated by periods of free motional evolution. This requires generalising the SDK unitary to allow multiple ions to be addressed within a single pulse. In this section, I first develop this generalisation and then express the resulting unitary gate evolution as a product of spin-dependent displacements and accumulated pairwise spin–spin phases, reflecting the form of the two-qubit case. This identifies the motional restoration and entangling-phase conditions required to implement the ideal operation stated in § 3.1.

3.2.1 Simultaneous state-dependent kicks for multiple ions

A single-ion SDK imparts a state-dependent momentum transfer to an addressed ion, exciting the collective motional modes of the chain. To implement a multi-qubit gate, the SDK formalism must therefore be extended to a set S containing $|S|$ distinct ions. The combined operation is constructed from the product of the single-ion SDKs acting on ion j defined in § 2.6.1,

$$\hat{U}_{\mathrm{SDK}}^{\mathrm{MQ}}(t) = \prod_{j \in S} \hat{U}_{\mathrm{SDK}}^{(j)}(t). \tag{3.3}$$

Although the ions are coupled through their collective motion, the SDK operators commute at equal times. Specifically, $[\hat{x}^{(j)}(t), \hat{x}^{(\ell)}(t)] = 0$, while spin operators on distinct ions commute with one another and with the position operators. Consequently, $[\hat{x}^{(j)}(t)\hat{\sigma}_z^{(j)}, \hat{x}^{(\ell)}(t)\hat{\sigma}_z^{(\ell)}] = 0$, allowing the product to be combined into a single exponen-

tial,

$$\hat{U}_{\mathrm{SDK}}^{\mathrm{MQ}}(t) = \exp\left[2ik\sum_{j\in S} z\hat{x}^{(j)}(t)\hat{\sigma}_z^{(j)}\right]. \tag{3.4}$$

This unitary transfers momentum $\pm 2z\hbar k$ to each addressed ion, conditioned on its internal state, while leaving the internal-state populations unchanged. Although the SDKs act independently at the instant of application, the resulting state-dependent displacements evolve through the shared motional modes. Sequences of such kicks can therefore generate both the desired entangling phases within target pairs and crosstalk between them. Equation (3.4) thus provides the elementary building block for constructing the parallel fast-gate evolution.

3.2.2 Multi-qubit fast gate unitary

A multi-qubit fast gate is constructed from a time-ordered sequence of SDK groups separated by free motional evolution. Each group of SDKs is applied at time t_n with signed integer amplitude z_n, so that the gate scheme is specified by $\{z_n, t_n\}$. The resulting gate unitary, expressed in the interaction picture with respect to the free motional evolution, is

$$\hat{U}_G^{\mathrm{MQ}}[\{z_n,t_n\}] = \mathcal{T}\prod_{n=1}^{K}\hat{U}_{\mathrm{SDK}}^{\mathrm{MQ}}(t_n) = \mathcal{T}\prod_{n=1}^{K}\exp\left[2iz_nk\sum_{j\in S}\hat{x}^{(j)}(t_n)\hat{\sigma}_z^{(j)}\right]. \tag{3.5}$$

Here, K is the number of pulse groups and $\mathcal{T}$ places later SDK groups to the left, while free motional evolution is incorporated through the time dependence of $\hat{x}^{(j)}(t_n)$. Expanding the ion displacement in the normal-mode basis, as is done in Eq. (2.23), gives

$$\hat{x}^{(j)}(t) = \sum_{\alpha} b_{\alpha}^{(j)}\sqrt{\frac{\hbar}{2m\omega_{\alpha}}}\left(\hat{a}_{\alpha}e^{-i\omega_{\alpha}t} + \hat{a}_{\alpha}^{\dagger}e^{i\omega_{\alpha}t}\right), \tag{3.6}$$

where $b_{\alpha}^{(j)}$ is the participation of ion j in mode α, with frequency ω_{α}. Using the mode-dependent Lamb–Dicke parameter defined in Eq. (2.32), the gate unitary becomes

$$\hat{U}_G^{\mathrm{MQ}}[\{z_n,t_n\}] = \mathcal{T}\prod_{n=1}^{K}\exp\left[\sum_{\alpha}\left(\hat{\lambda}_{\alpha,n}\hat{a}_{\alpha}^{\dagger} - \hat{\lambda}_{\alpha,n}^{\dagger}\hat{a}_{\alpha}\right)\right], \tag{3.7}$$

where $\hat{\lambda}_{\alpha,n} = 2iz_n\eta_\alpha e^{i\omega_\alpha t_n}\sum_{j\in S} b_\alpha^{(j)}\hat{\sigma}_z^{(j)}$. This expresses each SDK group as a product of spin-dependent displacements of the independent motional modes.

Although SDK operators acting on different ions commute at equal times, successive SDK groups generally do not commute because the motional modes evolve between kicks. Consequently, their time-ordered product generates both a net displacement and an additional spin-dependent phase. Defining $\hat{A}_n = \sum_\alpha(\hat{\lambda}_{\alpha,n}\hat{a}_\alpha^\dagger - \hat{\lambda}_{\alpha,n}^\dagger\hat{a}_\alpha)$, the Baker–Campbell–Hausdorff (BCH) formula gives

$$\hat{U}_G^{\mathrm{MQ}} = \exp\left[\sum_{n=1}^{K}\hat{A}_n + \frac{1}{2}\sum_{m<n}[\hat{A}_n, \hat{A}_m]\right], \tag{3.8}$$

where, as shown below, all higher-order nested commutators vanish. The commutator between SDK groups is evaluated using the bosonic commutation relations $[\hat{a}_\alpha, \hat{a}_\beta^\dagger] = \delta_{\alpha\beta}$, where $\delta_{\alpha\beta}$ is the Kronecker delta. These relations follow from the canonical commutation of position and momentum operators and express the non-commutativity of phonon creation and annihilation, while operators belonging to different normal modes commute. Applying these relations gives

$$\begin{aligned}[\hat{A}_n, \hat{A}_m] &= \sum_\alpha\left(\hat{\lambda}_{\alpha,n}\hat{\lambda}_{\alpha,m}^\dagger - \hat{\lambda}_{\alpha,n}^\dagger\hat{\lambda}_{\alpha,m}\right) \\ &= 8iz_mz_n\sum_\alpha\eta_\alpha^2\sin[\omega_\alpha(t_n - t_m)]\left(\sum_{j\in S} b_\alpha^{(j)}\hat{\sigma}_z^{(j)}\right)^2.\end{aligned} \tag{3.9}$$

Importantly, this commutator contains only products of $\hat{\sigma}_z$ operators, with no remaining motional operators. Since these spin operators commute with one another and with the motional operators, $[[\hat{A}_n, \hat{A}_m], \hat{A}_r] = 0$ for all n, m, r. Consequently, all higher-order nested commutators vanish, making the BCH expansion exact. Physically, the non-zero commutator represents the spin-dependent geometric phase accumulated through successive motional displacements, which can remain non-zero even when the motional restoration conditions are satisfied.

Expanding the squared spin sum in Eq. (3.9), the diagonal terms contribute only a global phase because $(\hat{\sigma}_z^{(j)})^2 = \hat{\mathbb{1}}$, while the remaining terms generate pairwise spin–spin phases between the addressed ions. The gate unitary can therefore be factorised, up to a

global phase, into

$$\hat{U}_G^{\mathrm{MQ}} = \hat{U}_2 \hat{U}_1, \tag{3.10}$$

where $\hat{U}_2$ describes the residual spin-dependent motional displacements,

$$\hat{U}_2 = \prod_{\alpha} \hat{D}_\alpha(\hat{\Lambda}_\alpha), \tag{3.11}$$

with

$$\hat{\Lambda}_\alpha = 2i\eta_\alpha \sum_{n=1}^{K} z_n e^{i\omega_\alpha t_n} \sum_{j \in S} b_\alpha^{(j)} \hat{\sigma}_z^{(j)}. \tag{3.12}$$

Here, $\hat{D}_\alpha(\hat{\Lambda}_\alpha)$ is the displacement operator for mode α, defined as $\hat{D}_\alpha(\hat{\Lambda}_\alpha) = \exp(\hat{\Lambda}_\alpha \hat{a}_\alpha^\dagger - \hat{\Lambda}_\alpha^\dagger \hat{a}_\alpha)$. The operator $\hat{U}_2$ therefore describes the residual entanglement between the internal and motional states following the SDK sequence. For ideal motional restoration, $\hat{\Lambda}_\alpha = 0$ for every mode, ensuring that the motional trajectories close and the internal states disentangle from the motion, in this case $\hat{U}_2 = \hat{\mathbb{1}}$.

The second contribution, $\hat{U}_1$, describes the accumulated spin–spin interactions,

$$\hat{U}_1 = \exp\left[i \sum_{j<\ell \in S} \Theta_{j\ell} \hat{\sigma}_z^{(j)} \hat{\sigma}_z^{(\ell)} \right], \tag{3.13}$$

where

$$\Theta_{j\ell} = 8 \sum_{\alpha} \eta_\alpha^2 b_\alpha^{(j)} b_\alpha^{(\ell)} \sum_{m<n} z_m z_n \sin[\omega_\alpha (t_n - t_m)]. \tag{3.14}$$

Unlike $\hat{U}_2$, this contribution acts exclusively on the internal states and generates the entangling phases required for multi-qubit entangling gates. Crucially, Eq. (3.14) generates phases between all addressed ion pairs, including those not intended to interact. Hence, crosstalk must be suppressed while preserving the desired entangling phases within each target pair.

Equation (3.10) therefore identifies the two conditions necessary for high-fidelity parallel gates: restoration of the collective motional modes and accumulation of the desired pairwise phases without crosstalk. The expressions above assume a common SDK sequence across all addressed ions. I next generalise the derivation to allow independently controlled SDK amplitudes and timings for each target pair.

3.2.3 Independent pulse sequences

To describe independently driven pairs, I collect all pulse times into a common ordered list $\{t_n\}$ and assign an integer amplitude $z_n^{(j)}$ to ion j at each time, with $z_n^{(j)} = 0$ when that ion is not addressed. The gate unitary then becomes

$$\hat{U}_G^{\mathrm{MQ}} = \mathcal{T} \prod_{n=1}^{K} \exp\left[2ik \sum_{j\in S} z_n^{(j)} \hat{x}^{(j)}(t_n) \hat{\sigma}_z^{(j)}\right]. \tag{3.15}$$

The same BCH construction gives

$$\hat{\Lambda}_\alpha = 2i\eta_\alpha \sum_{j\in S} b_\alpha^{(j)} \hat{\sigma}_z^{(j)} \sum_{n=1}^{K} z_n^{(j)} e^{i\omega_\alpha t_n}, \tag{3.16}$$

and

$$\Theta_{j\ell} = 4 \sum_\alpha \eta_\alpha^2 b_\alpha^{(j)} b_\alpha^{(\ell)} \sum_{m<n} \left[z_n^{(j)} z_m^{(\ell)} + z_n^{(\ell)} z_m^{(j)}\right] \sin[\omega_\alpha (t_n - t_m)]. \tag{3.17}$$

Setting $z_n^{(j)} = z_n$ for every $j \in S$ recovers Eqs. (3.12) and (3.14), i.e. the common-sequence expressions. Within each target pair, both ions receive the same SDK strengths at the same times. However, the strengths and timings applied to (i, j) can differ from those applied to (k, ℓ) – this will be the basis for the parallel gate protocols I develop in the following chapters. I make this choice instead of assigning each ion its own pulse sequence to reduce the degrees of freedom in the optimisation procedure. Moreover, since each sequence is designed to implement an entangling operation on its respective ion pair, the resulting sequences can also be applied independently as conventional two-qubit gates. This is discussed further in § 4.1.

Independent addressing preserves the separation of the gate evolution into residual displacements and pairwise spin–spin phases. To enact the ideal operation defined in § 3.1, the motion of the ions must be restored at the end of the sequence and each pair phase must equal its target value. For parallel two-qubit gates, this requires the desired entangling phases on the target pairs and zero crosstalk. The state-averaged fidelity will be applied to the multi-qubit case to quantify whether these conditions are met.

3.3 Multi-qubit gate fidelity

With the ideal and physical gate unitaries defined in Eqs. (3.1) and (3.10), respectively, I now quantify gate performance using the state-averaged fidelity introduced in § 2.1.2. As trapped-ion gates operate through collective motional excitation, not restoring this motion by the end of the gate can leave the internal states entangled with the motion, reducing the fidelity. The resulting spin evolution is therefore obtained by tracing over the motional degrees of freedom, assumed to be initially in a thermal state ρ_{th}. The state-averaged fidelity can then be evaluated using the operator-basis formula derived by Nielsen [36],

$$\mathcal{F} = \frac{\sum_l \mathrm{Tr}\left[\hat{U}_{\mathrm{ideal}} \hat{W}_l^\dagger \hat{U}_{\mathrm{ideal}}^\dagger \, \mathrm{Tr}_m\left[\hat{U}_G \left(\hat{W}_l \otimes \rho_{\mathrm{th}}\right) \hat{U}_G^\dagger\right]\right] + d^2}{d^2(d+1)}, \tag{3.18}$$

where $d = 2^{2M}$ is the dimension of the Hilbert space with M the number of parallel two-qubit gates acting on distinct ion pairs. Here, Tr_m denotes the partial trace over motion, while the outer trace is over the spin degrees of freedom. The operators $\{\hat{W}_l\}$ form a complete orthogonal basis of $d \times d$ matrices, normalised such that $\mathrm{Tr}(\hat{W}_k^\dagger \hat{W}_l) = d\delta_{kl}$. A convenient choice is the set of Pauli strings (recall the definition of the Pauli matrices in § 2.1.2),

$$\{\hat{W}_l\} = \left\{\hat{\sigma}_{\mu_1}^{(j_1)} \otimes \cdots \otimes \hat{\sigma}_{\mu_{2M}}^{(j_{2M})} : \mu_r \in \{0, x, y, z\}\right\}, \tag{3.19}$$

where $\hat{\sigma}_0 = \hat{\mathbb{1}}$ and the superscript j_r identifies the ion on which each operator acts. This basis allows the fidelity to be evaluated over all spin configurations without explicitly averaging over initial states as is done in previous fast gate theory [60, 67]. Explicitly averaging over all initial states is not feasible in the multi-qubit case as it becomes quickly intractable with the number of addressed ions. As quantum computation requires small errors ($< 10^{-2}$) for error correcting algorithms [38], we exclusively work in the small-error regime, where the leading contributions to infidelity arise from residual spin–motion entanglement and deviations from the desired entangling phases. These contributions are additive to leading quadratic order and can therefore be evaluated separately in the following subsections, significantly simplifying the computation.

3.3.1 Motional infidelity

The motional contribution quantifies the reduction in fidelity due to incomplete restoration of the collective motional modes. Starting from the factorisation $\hat{U}_G^{\mathrm{MQ}} = \hat{U}_2\hat{U}_1$ in Eq. (3.10), I isolate this contribution by setting $\hat{U}_1 = \hat{U}_{\mathrm{ideal}}$. Physically, this assumes that the desired entangling phases are accumulated exactly and that crosstalk is suppressed, so that any remaining infidelity arises solely from improper restoration of the motional modes. Since $\hat{U}_1 = \hat{U}_{\mathrm{ideal}}$ acts only on the spins and commutes with $\hat{U}_2$, it can be taken outside the partial trace over motion. The ideal unitary factors then cancel by the cyclic property of the trace, reducing Eq. (3.18) to

$$\mathcal{F}_{\mathrm{mot}} = \frac{\sum_l \mathrm{Tr}\left[\hat{W}_l^\dagger \,\mathrm{Tr}_m\left[\hat{U}_2\left(\hat{W}_l \otimes \rho_{\mathrm{th}}\right)\hat{U}_2^\dagger\right]\right] + d^2}{d^2(d+1)}. \tag{3.20}$$

To evaluate the motional trace, I consider its action on a computational basis operator $|s\rangle\langle s'|$. Each spin configuration s is specified by the Pauli-z eigenvalues $s_j \in \{-1, +1\}$, such that $\hat{\sigma}_z^{(j)}|s\rangle = s_j|s\rangle$, with s and s' independently ranging over all d configurations. Since $\hat{U}_2$ is diagonal in the computational spin basis, each configuration produces a corresponding displacement $\Lambda_\alpha(s)$ of the motional mode α. Different spin configurations can therefore leave the motional modes in different states. Replacing each $\hat{\sigma}_z^{(j)}$ with its eigenvalue gives the mode displacement,

$$\Lambda_\alpha(s) = 2i\eta_\alpha \sum_{j\in S} b_\alpha^{(j)} s_j \sum_{n=1}^{K} z_n^{(j)} e^{i\omega_\alpha t_n} = \sum_{j\in S} C_{\alpha j} s_j, \tag{3.21}$$

where $C_{\alpha j} = 2i\eta_\alpha b_\alpha^{(j)} \sum_{n=1}^{K} z_n^{(j)} e^{i\omega_\alpha t_n}$ is the residual displacement amplitude associated with ion j in mode α. The total displacement therefore depends on the spin configuration, with perfect motional restoration requiring $C_{\alpha j} = 0$ for every addressed ion and mode.

With this, I can now evaluate the motional trace using the displacement operator identities $\hat{D}(\beta)\hat{D}(\gamma) = \exp[(\beta\gamma^* - \beta^*\gamma)/2]\hat{D}(\beta+\gamma)$ and $\mathrm{Tr}[\hat{D}_\alpha(\beta)\rho_{\mathrm{th},\alpha}] = \exp[-|\beta|^2(\bar{n}_\alpha + \frac{1}{2})]$,

$$\mathrm{Tr}_m\left[\hat{U}_2\left(|s\rangle\langle s'| \otimes \rho_{\mathrm{th}}\right)\hat{U}_2^\dagger\right] = \Gamma_{ss'}|s\rangle\langle s'|, \tag{3.22}$$

where I define the coherence factor to be,

$$\Gamma_{ss'} = \exp\left[-\sum_{\alpha} |\Lambda_\alpha(s) - \Lambda_\alpha(s')|^2 \left(\bar{n}_\alpha + \frac{1}{2}\right)\right], \tag{3.23}$$

with $\bar{n}_\alpha = [\exp(\hbar\omega_\alpha/k_B T) - 1]^{-1}$ the mean thermal occupation of mode α. By linearity, the result extends to any basis operator $\hat{W}_l = \sum_{s,s'}(W_l)_{ss'}|s\rangle\langle s'|$, giving

$$\mathcal{F}_{\text{mot}} = \frac{\sum_{l,s,s'} |(W_l)_{ss'}|^2 \Gamma_{ss'} + d^2}{d^2(d+1)}. \tag{3.24}$$

Completeness of the orthogonal operator basis, together with its normalisation, implies $\sum_l |(W_l)_{ss'}|^2 = d$. Consequently,

$$1 - \mathcal{F}_{\text{mot}} = \frac{1}{d(d+1)} \sum_{s,s'} (1 - \Gamma_{ss'}) \simeq \frac{1}{d(d+1)} \sum_{\alpha} \left(\bar{n}_\alpha + \frac{1}{2}\right) \sum_{s,s'} |\Lambda_\alpha(s) - \Lambda_\alpha(s')|^2, \tag{3.25}$$

where the approximation follows by expanding the coherence factor for small residual displacements. The leading infidelity is therefore quadratic in the differences between the mode displacements, $\Lambda_\alpha(s)$.

To evaluate the remaining spin sum, each eigenvalue $s_j = \pm 1$ occurs equally often across the complete computational basis, giving $\sum_s s_j = 0$ and $\sum_s s_j s_k = d\delta_{jk}$. Summing independently over the two spin configurations yields $\sum_{s,s'}(s_j - s'_j)(s_k - s'_k) = 2d^2\delta_{jk}$. Substituting the mode displacements gives

$$\sum_{s,s'} |\Lambda_\alpha(s) - \Lambda_\alpha(s')|^2 = \sum_{j,k \in S} C_{\alpha j} C^*_{\alpha k} \sum_{s,s'} (s_j - s'_j)(s_k - s'_k) = 2d^2 \sum_{j \in S} |C_{\alpha j}|^2. \tag{3.26}$$

Thus, the cross terms between different ions vanish upon averaging over the complete spin basis, leaving a sum of their individual contributions to the residual motional infidelity. Substituting $C_{\alpha j}$ and $d = 2^{2M}$ gives

$$1 - \mathcal{F}_{\text{mot}} \simeq \frac{2^{2M}}{2^{2M}+1} \sum_{\alpha} (1 + 2\bar{n}_\alpha)\, 4\eta_\alpha^2 \sum_{j \in S} \left| \sum_{n=1}^{K} z_n^{(j)} e^{i\omega_\alpha t_n} \right|^2 \left(b_\alpha^{(j)}\right)^2. \tag{3.27}$$

This expression quantifies the leading infidelity arising from incomplete motional restoration, weighted by the thermal occupation and participation of each mode. Importantly,

the motional infidelity vanishes when all conditional displacements close, independently of the accumulated entangling phases. This establishes the motional restoration requirement, while deviations from the desired entangling phases are considered separately in the following subsection.

3.3.2 Phase infidelity

The phase contribution quantifies errors in the accumulated spin–spin interactions, even when the collective motional modes are perfectly restored. To isolate this contribution, I assume perfect motional restoration by setting $\hat{U}_2 = \hat{\mathbb{1}}$, such that the gate unitary becomes $\hat{U}_G^{\mathrm{MQ}} = \hat{U}_1$. Since $\hat{U}_1$ acts only on spin states and $\mathrm{Tr}_m(\rho_{\mathrm{th}}) = 1$, the motional trace in Eq. (3.18) becomes trivial. Therefore, the infidelity arises from deviations in the desired entangling phases or crosstalk between target pairs. To proceed, I define the phase-error unitary as

$$\hat{U}_{\mathrm{phase}} = \hat{U}_{\mathrm{ideal}}^{\dagger}\hat{U}_1 = \exp\left[i\sum_{j<\ell}\Delta\Theta_{j\ell}\hat{\sigma}_z^{(j)}\hat{\sigma}_z^{(\ell)}\right], \tag{3.28}$$

where $\Delta\Theta_{j\ell} = \Theta_{j\ell} - \Theta_{j\ell}^{\mathrm{tar}}$ is the deviation from the target phase for each ion pair. Thus, $\hat{U}_{\mathrm{phase}}$ describes the residual spin evolution relative to the ideal gate, with $\hat{U}_{\mathrm{phase}} = \hat{\mathbb{1}}$ corresponding to perfect phase accumulation. Using that the trace is cyclic, the fidelity in Eq. (3.18) becomes

$$\mathcal{F}_{\mathrm{phase}} = \frac{\sum_l \mathrm{Tr}\left[\hat{W}_l^{\dagger}\hat{U}_{\mathrm{phase}}\hat{W}_l\hat{U}_{\mathrm{phase}}^{\dagger}\right] + d^2}{d^2(d+1)}, \tag{3.29}$$

with $d = 2^{2M}$. Completeness of the operator basis, normalised by $\mathrm{Tr}(\hat{W}_k^{\dagger}\hat{W}_l) = d\delta_{kl}$, gives $\sum_l \hat{W}_l^{\dagger}\hat{A}\hat{W}_l = d\,\mathrm{Tr}(\hat{A})\hat{I}$ for any spin operator $\hat{A}$. Applying this identity with $\hat{A} = \hat{U}_{\mathrm{phase}}$ reduces the operator-basis sum to $d|\,\mathrm{Tr}(\hat{U}_{\mathrm{phase}})|^2$, yielding

$$1 - \mathcal{F}_{\mathrm{phase}} = \frac{d^2 - \left|\mathrm{Tr}(\hat{U}_{\mathrm{phase}})\right|^2}{d(d+1)}. \tag{3.30}$$

This is the exact state-averaged infidelity of the phase-error unitary [36].

Since $\hat{U}_{\mathrm{phase}}$ is diagonal in the computational spin basis, each spin configuration ac-

quires a phase determined by the pairwise phase errors. Using the spin-eigenvalue labelling introduced above, its trace can be evaluated directly as

$$\mathrm{Tr}(\hat{U}_{\mathrm{phase}}) = \sum_{s\in\{\pm1\}^{2M}} \exp\left[i\sum_{j<\ell}\Delta\Theta_{j\ell}s_j s_\ell\right]. \tag{3.31}$$

This expression sums the phase factors acquired by all computational basis states. When these phases are identical, they contribute only a global phase and the fidelity remains unity. Conversely, differences between the accumulated phases reduce the magnitude of the trace, reflecting the incorrect relative phases introduced between spin configurations. The phase infidelity admits an exact finite-sum expression suitable for numerical optimisation, although the number of terms grows exponentially with M.

A simpler analytical expression follows in the small-error regime. Writing $\hat{U}_{\mathrm{phase}} = e^{i\hat{H}}$, where $\hat{H} = \sum_{j<\ell}\Delta\Theta_{j\ell}\hat{\sigma}_z^{(j)}\hat{\sigma}_z^{(\ell)}$, tracelessness and orthogonality of the distinct Pauli strings imply $\mathrm{Tr}(\hat{H}) = 0$ and $\mathrm{Tr}(\hat{H}^2) = d\sum_{j<\ell}\Delta\Theta_{j\ell}^2$. Expanding the trace to second order and substituting $d = 2^{2M}$ therefore gives

$$1 - \mathcal{F}_{\mathrm{phase}} \simeq \frac{\mathrm{Tr}(\hat{H}^2)}{d+1} = \frac{2^{2M}}{2^{2M}+1}\sum_{j<\ell}|\Delta\Theta_{j\ell}|^2. \tag{3.32}$$

The orthogonality of the Pauli strings eliminates cross terms between different ion pairs at quadratic order, leaving an additive contribution from each phase error. Importantly, the expression treats deviations in the desired entangling phases and crosstalk with the same weighting. Consequently, even if the target pairs acquire their correct entangling phases, crosstalk can substantially reduce the parallel-gate fidelity.

3.4 Implications of multi-qubit fast gate theory

In this chapter, the fidelity for the two-qubit fast entangling gate formalism was generalised to the case of multi-qubit gates with a particular focus on parallel gate implementation. The starting point was defining the ideal multi-qubit gate unitary in § 3.1 from which the physical multi-qubit gate unitary was constructed using a sequence of time ordered multi-qubit SDKs in § 3.2.2. With a notion of the ideal and physical gate

unitaries defined, I was able to derive a state averaged fidelity, and subsequent infidelity for both the motional and phase contribution in § 3.3. In the small error regime the contributions are independent and can therefore be summed to give,

$$1-\mathcal{F} \simeq \frac{2^{2M}}{2^{2M}+1}\left(\sum_{j<\ell}|\Delta\Theta_{j\ell}|^2+\sum_{\alpha}(1+2\bar{n}_\alpha)\,4\eta_\alpha^2\sum_{j\in S}\left|\sum_{n=1}^{K}z_n^{(j)}e^{i\omega_\alpha t_n}\right|^2\left(b_\alpha^{(j)}\right)^2\right). \tag{3.33}$$

Therefore, to minimise the infidelity a gate scheme must simultaneously ensure that $|\Delta\Theta_{j\ell}|^2$ and $C_{\alpha j}\forall\alpha, j$ are as close to zero as possible. Physically, $|\Delta\Theta_{j\ell}|^2$ being small corresponds to accumulating the desired entangling phase within pairs and keep entangling phases between undesired pairs small. The second condition of ensuring that $C_{\alpha j}=0,\forall\alpha,j$ is small is instead the condition that each motional mode is restored by the end of the gate. These are therefore the multi-qubit extension of the condition equations (2.45) and (2.44) introduced in § 2.6.2. A careful examination of Eq. (3.33) reveals that in the case of $M=1$ (i.e. a two-qubit gate) the prefactor is $\frac{2^2}{2^2+1}=\frac{4}{5}$ which is not equal to what is displayed in Eq. (2.46). The reason for this is because here the infidelity is not found by explicitly averaging over all possible states and so the choice of measure is different to Refs. [60, 67]. However, this difference is only at the level of the prefactor and does not change the description of the physics, nor the utility of Eq. (3.33) in this thesis.

From the form of Eq. (3.33), we can infer the increased difficulty of numerically optimising for a bespoke parallel gate. For a two-qubit gate in a chain of N ions, there are N motional restoration conditions that must be satisfied, one for each motional mode and only one phase condition. In a parallel gate, the same pulse sequence generates phases between every pair of addressed ions. Therefore, in addition to producing the desired phases on the M target pairs, the optimisation must suppress crosstalk. For $2M$ addressed ions, there are $\binom{2M}{2}$ pairwise phases in total, of which only M are desired. Thus the number of crosstalk terms grows as

$$\binom{2M}{2}-M=2M(M-1). \tag{3.34}$$

This rapidly, $\mathcal{O}(M^2)$, increases the number of phase constraints that must be satisfied as

more gates are implemented in parallel. With this quadratic scaling, it suggests that numerically optimising many two-qubit gates in parallel will be a computationally intensive procedure relative to previous fast gate optimisation. The following numerical analysis investigates how closely the motional and phase conditions can be satisfied in the context of parallel gate optimisation.

Chapter 4

Naive parallelisation of two-qubit fast gates

In Chapter 3, I extended the well-understood theory of two-qubit fast entangling gates to describe the multi-qubit operations studied in this thesis. In doing so, I identified crosstalk as an additional error source that must be considered when implementing parallel two-qubit gates. Suppressing this crosstalk during numerical optimisation would increase the complexity of searching for parallel solutions. Therefore, rather than immediately using the multi-qubit infidelity derived in Chapter 3 as a cost function for numerical searches, I first use it to assess a computationally efficient approach termed *naive parallelisation*.

In this chapter, I investigate whether existing two-qubit fast-gate solutions can be parallelised by simultaneously applying two independently optimised gate schemes to disjoint target pairs. I define this protocol in § 4.1 and assess its performance across representative pair geometries in § 4.2. The physical mechanisms responsible for the success or failure of naive parallelisation are then examined in § 4.3. The impact of the spatial separation of the target pairs and the introduction of a timing delay between the two gates is analysed in § 4.4. The main findings of this chapter are summarised in § 4.5.

4.1 Methods for naive parallelisation

4.1.1 Naive parallelisation protocol

The simplest way to construct a parallel fast gate is to simultaneously apply two pulse sequences that have already been optimised for independent two-qubit gates, without further optimisation. This is what I define as *naive parallelisation*. Mathematically,

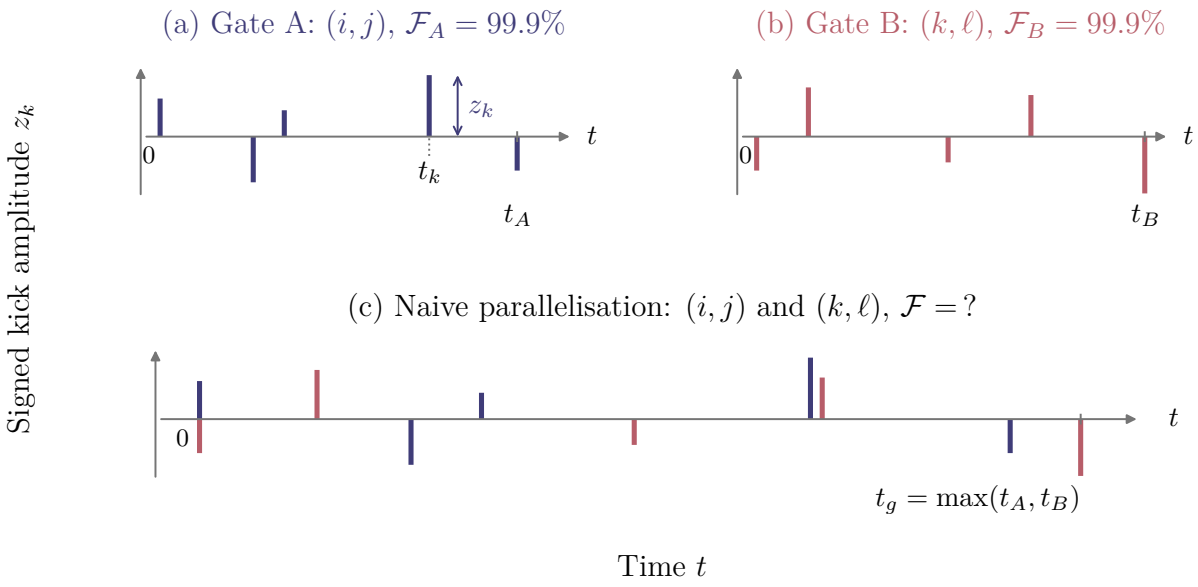


Fig. 4.1: A schematic of the naive parallelisation protocol for (a) gate A entangling ions (i,j) and (b) gate B entangling ions (k,ℓ). Each pulse group has arrival time t_k and signed SDK strength z_k. (c) A demonstration of how one takes both gate schemes and enacts them simultaneously. The timings and SDK strengths are invariant under this protocol with the parallel gate duration defined as the longest of the two constituent gates $t_g = \max(t_A, t_B)$. Whilst the fidelities of gates A and B are at least 99.9%, this does not guarantee that the naive protocol is capable of producing an equally high-fidelity gates due to crosstalk and hence the fidelity of the parallel operation is denoted as '$\mathcal{F} = ?$'. Pulse sequences are illustrative and typically contain many (> 32) pulse groups.

consider two gate schemes, $\{z_n^{(ij)}, t_n^{(ij)}\}$ and $\{z_m^{(k\ell)}, t_m^{(k\ell)}\}$, acting on disjoint ion pairs (i,j) and (k,ℓ), respectively. The naively parallel gate scheme is constructed by combining their pulse sequences in chronological order, retaining each pulse group's amplitude, application time and target-pair assignment. Coincident pulse groups are applied simultaneously with both gates enforced to start at $t = 0$, as illustrated in Fig. 4.1.

The quality of the combined operation is evaluated using the multi-qubit infidelity derived in Chapter 3, which, assuming small errors, comprises three contributions from restoring motional modes ϵ_{motion}, acquiring the desired entangling phase, ϵ_{target} and crosstalk $\epsilon_{\text{crosstalk}}$. However, as the analysis in § 3.3 is done for both motional and phase fidelity separately, I combine crosstalk and entangling phase errors as phase fidelity in the following analysis. Crosstalk represents the additional error arising from unwanted entangling phases between the two target pairs which is a unique property of multi-qubit entanglement. For example, in the case of two two-qubit gates in parallel, there are six

target entangling-phase conditions,

$$\Theta_{ij}^{\text{tar}} = \Theta_{k\ell}^{\text{tar}} = \frac{\pi}{4}, \qquad \Theta_{pq}^{\text{tar}} = 0 \quad \forall\, p \in \{i,j\},\ q \in \{k,\ell\}. \tag{4.1}$$

Importantly, the residual motional displacement is linear in the applied pulse sequences. Consequently, the displacement of mode α for the combined operation is

$$\hat{\Lambda}_{\alpha}^{(ij,k\ell)} = \hat{\Lambda}_{\alpha}^{(ij)} + \hat{\Lambda}_{\alpha}^{(k\ell)}. \tag{4.2}$$

If both constituent gates independently restore the motion, their combined displacement must also vanish. Conversely, since the two contributions act on disjoint sets of spin operators, which are linearly independent, vanishing total displacement requires each contribution to vanish individually. Furthermore, the target-pair phases remain unchanged because each sequence acts only on its original pair. Thus, satisfying the motional-restoration and target-phase conditions for the parallel operation is equivalent to satisfying them for both constituent gates independently.

However, whilst this argument is true in the two-qubit case it does not follow for the multi-qubit case. Mixed terms between the two sequences generate crosstalk. Since the individual gate optimisations do not constrain these mixed terms, even high-fidelity constituent gates can produce a parallel operation with nonzero crosstalk.

4.1.2 Target-pair geometries

Even when considering only two simultaneous two-qubit gates, there are $\frac{1}{2}\binom{N}{2}\binom{N-2}{2}$ ways to choose two disjoint pairs in an N-ion chain, corresponding to 630 configurations for $N = 10$. Studying all possible geometries is therefore impractical. To investigate how pair separation and gate time regime influence parallel gate fidelity, I restrict the analysis to four representative configurations, shown in Fig. 4.2.

The first two geometries contain nearest-neighbour pairs. In the first, the pairs are well separated along the chain, whereas in the second, they are adjacent. Nearest-neighbour gates can operate in the supersonic regime, where the driven motion remains concentrated near the addressed ions [29]. I therefore expect well-separated pairs to experience

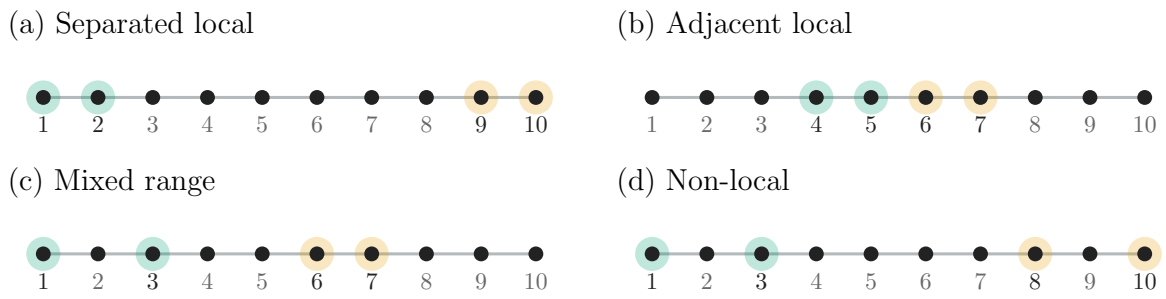


Fig. 4.2: Representative target-pair geometries in a ten-ion chain that are considered in this analysis. Green and yellow identify the two independently addressed ion pairs. The other ions are unaddressed by the SDKs. The cases considered are (a) well-separated local pairs $(1, 2)$ and $(9, 10)$, (b) adjacent local pairs $(4, 5)$ and $(6, 7)$, (c) a non-local pair $(1, 3)$ alongside a local pair $(6, 7)$, and (d) two non-local pairs $(1, 3)$ and $(8, 10)$.

negligible crosstalk. By comparing this to the case of adjacent nearest-neighbour pairs, I can assess the influence of pair separation.

I also consider non-nearest-neighbour pairs, where either two non-local gates or one non-local gate alongside a local gate are applied in parallel. Previous work has identified resource-efficient non-local fast gates in the subsonic regime, where collective motion propagates throughout the ion chain [29]. These configurations therefore allow me to investigate whether the non-localised motional dynamics associated with non-local interactions introduces greater crosstalk, and whether local gates can still be implemented alongside them with high fidelity.

Whilst there might exist pair geometries that are not strongly susceptible to crosstalk, strictly restricting parallel operations to these favourable configurations would limit the ability to exploit the all-to-all connectivity of trapped-ion processors. Indeed, during parallel operation a quantum algorithm may require a wide-range of pair geometries necessitating the study of the parallelisability of both local and non-local pair geometries.

4.1.3 Numerical parameters

I consider a linear chain of ten $^{133}Ba^{+}$ ions confined in a harmonic axial trapping potential with a centre-of-mass mode frequency of $\omega_z/(2\pi) \approx 411\,\text{kHz}$ with a total length of approximately $30\,\mu\text{m}$. The qubit states are encoded in the hyperfine manifold of the $6^2S_{1/2}$ ground state, with SDKs implemented using Raman transitions driven by $\lambda = 532\,\text{nm}$

light aligned with the trap axis, giving an effective wavevector $k = 4\pi/\lambda = 2k$. The SDKs couple exclusively to the axial motional modes, which are assumed to be initially in thermal states at a temperature of 30 µK. For the axial mode frequencies considered, this corresponds to a mode-averaged mean thermal phonon occupation of $\bar{n}_{\text{avg}} = 0.22$, indicating low average motional excitation. Such low thermal occupations are experimentally achievable through techniques including Raman sideband cooling, which has been demonstrated across the axial modes of chains containing up to 24 ions [76]. Gate durations are expressed in units of the characteristic propagation time, $t_{\text{char}} \simeq 2L/c_s$, where L is the length of the chain and c_s is the effective phonon propagation speed [29]. For the 10-ion chain considered here, one finds the characteristic propagation time to be $t_{\text{char}} = 3.32\,\mu\text{s}$.

Each constituent two-qubit gate uses 48 pulse groups, with at most five SDKs per group. This limits the pulse resources available to each gate, motivated by the accumulation of errors associated with imperfect SDKs [58, 68]. These imperfections are not explicitly included in the calculated infidelities but as SDK errors compound it is possible to instead minimise the total amount of SDKs as proxy. This is explored in detail in § 5.3.1.

Although finite SDK repetition rates are not considered in this analysis, previous two-qubit fast-gate studies have demonstrated that high-fidelity subsonic gates can be achieved at repetition rates as low as 25 MHz [68]. This suggests that the subsonic regime is compatible with experimentally accessible laser repetition rates, although this has not been explicitly verified for the parallel sequences considered here.

4.2 Naive parallelisation performance

Fig. 4.3 shows the infidelity for well-separated and adjacent local pairs in a ten-ion chain. For well-separated pairs, naive parallelisation gives an infidelity of approximately 10^{-5}–10^{-4} in the supersonic regime. As the gate time approaches t_{char}, the infidelity rises sharply to 10^{-1} or greater and remains phase dominated at longer times. Here, and for all infidelity plots presented in this thesis, phase infidelity refers to the contribution from both desired and crosstalk phases. This behaviour is consistent with motion generated

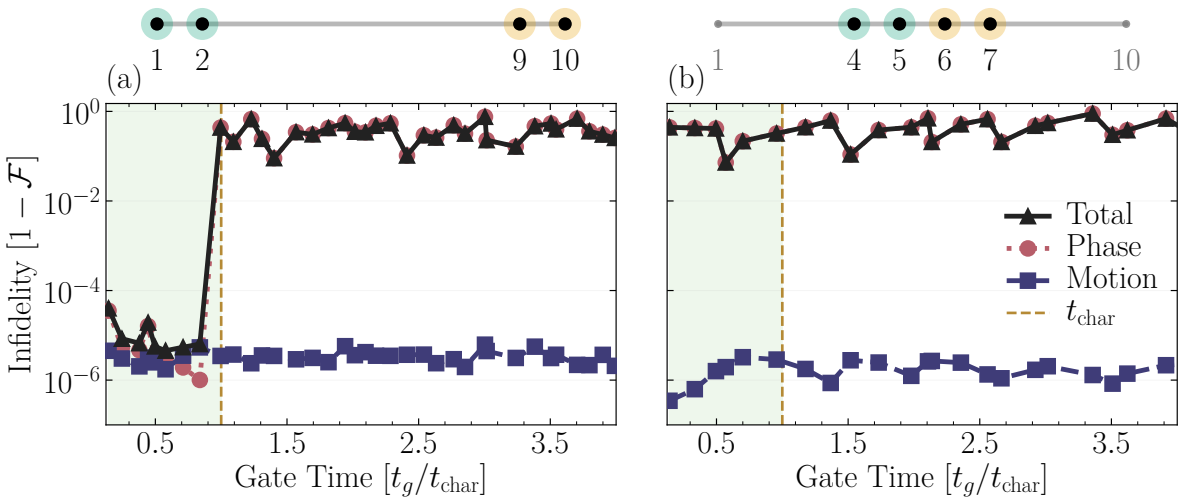


Fig. 4.3: State-averaged infidelity for two local gates in a ten-ion chain, applied to (a) well-separated pairs $(1,2)$ and $(9,10)$. Adjacent pairs $(4,5)$ and $(6,7)$ are shown in (b). Total infidelity and its phase (both target and crosstalk) and motional contributions are plotted against gate time normalised by the roundtrip time for phonon propagation in the chain, t_{char}. The shaded region denotes supersonic gate times, with the dashed line marking $t_g = t_{\text{char}}$. Well-separated pairs achieve low ($\sim 10^{-5}$) infidelity only when $t_g < t_{\text{char}}$, whereas adjacent pairs are phase limited throughout the range shown.

at one pair reaching the other as the gate duration approaches the characteristic phonon propagation time, $t_{\text{char}} \simeq 2L/c_s$, corresponding approximately to the time required for a disturbance to travel along the chain and return [29]. Therefore, gates that are in the subsonic regime ($t_g/t_{\text{char}} > 1$) may be phase dominated due to the motion between the pairs acting as a medium for crosstalk.

For adjacent pairs, the parallel gate is phase dominated across the entire range of gate times shown, with infidelities generally of order unity. The large difference in the parallel performance as compared to the serial application likely arises from crosstalk. I examine the error contribution due to crosstalk in § 4.3.

Fig. 4.4 shows the same comparison for a non-local gate applied alongside a local gate (a) and for two non-local gates applied together (b). Neither geometry reaches a parallel infidelity below 10^{-2} over the gate times shown. For both subsonic and supersonic gates, the motional infidelity remains much smaller than the phase contribution, while the total infidelity ranges between approximately 10^{-1} and unity. As with the well-separated local pairs, motional restoration at longer gate times does not guarantee a

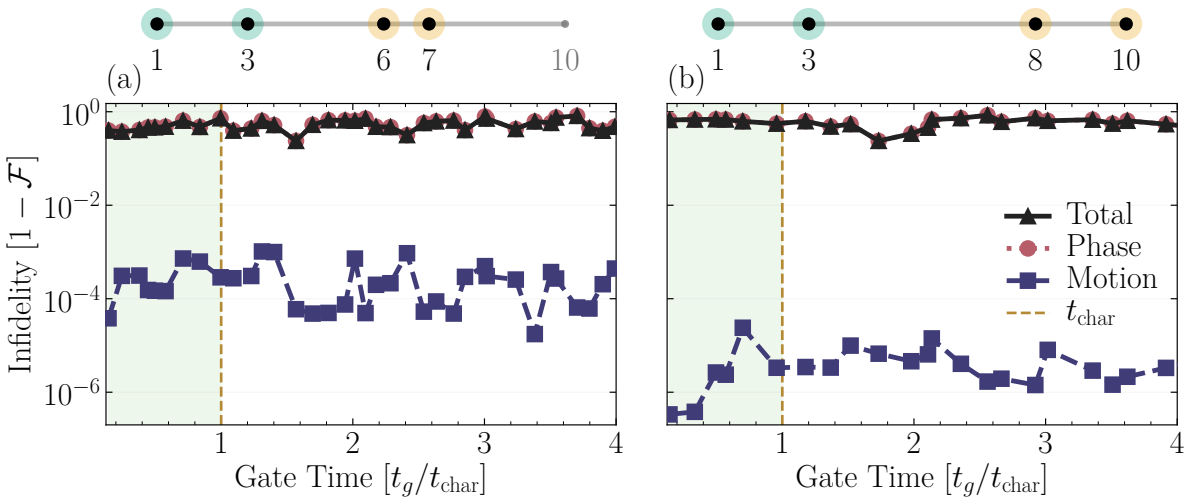


Fig. 4.4: Same as Fig. 4.3 but for (a) a non-local pair $(1,3)$ and a local pair $(6,7)$, and (b) two non-local pairs $(1,3)$ and $(8,10)$, in a ten-ion chain. Low infidelity over the gate times shown is never achieved due to dominant phase errors.

successful parallel gate. In § 4.1, I showed that although the naively parallel gates inherit low ϵ_{motion} and ϵ_{target}, the overall infidelity can be limited by $\epsilon_{\text{crosstalk}}$ due to cross terms between the gates. To understand the evolution of $\epsilon_{\text{crosstalk}}$, I examine its magnitude and time-evolution in § 4.3.

4.3 Crosstalk evolution in naively parallel gates

The results in § 4.2 show that phase error limits naive parallelisation in several of the geometries considered. To determine its source, I first distinguish errors in the intended entangling phases from crosstalk. I then examine when these phases accumulate as a result of the evolution of the ions motion during the gate. I focus on the well-separated local pairs because this geometry exhibits both a low-infidelity supersonic regime and a high-infidelity subsonic regime, allowing the two behaviours to be compared directly.

Fig. 4.5 shows the absolute difference between the realised and target phases defined in Eq. (4.1). For the target pairs $(1,2)$ and $(9,10)$, the target entangling phase is $\pi/4$, whereas the four crosstalk terms have a target of zero. A gate satisfying all phase conditions would exhibit $|\Delta\Theta_{j\ell}|/\pi = 0$ for all $j,\ell \in \{1,2,9,10\}$. In the supersonic regime (a), the errors in the desired entangling phases are of order $10^{-5}\pi$, while the crosstalk

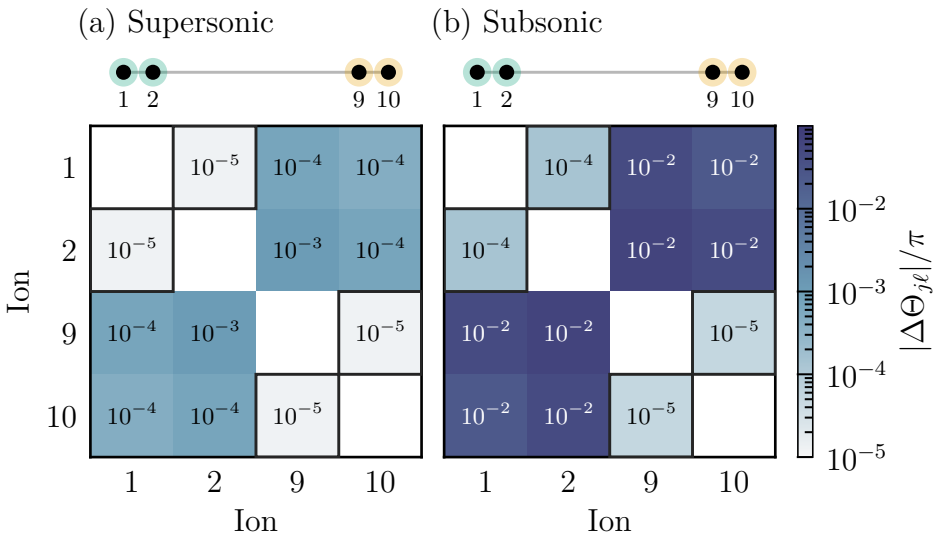


Fig. 4.5: Deviations from the target phases, $|\Delta\Theta_{j\ell}|/\pi$, for naive parallel gates on the well-separated ion pairs $(1, 2)$ and $(9, 10)$ in a ten-ion chain. The left panel shows a supersonic gate (t_g =1.48 µs) and the right panel a subsonic gate (t_g =4.65 µs). Outlined entries correspond to the desired entangling phases. The remaining off-diagonal entries represent crosstalk. Target-phase errors remain small in both cases, while crosstalk increases substantially in the subsonic gate. It should be noted that these matrices are symmetric and are shown in full for completeness.

is at most $10^{-3}\pi$. In the subsonic regime (b), the desired phase errors remain similarly small, but crosstalk grows to approximately 0.03π–0.07π. The large phase infidelity in this example therefore arises from crosstalk between the two intended gates, rather than a failure to accumulate their target entangling phases. Indeed, it should be expected that both cases have errors on the order of 10^{-4} given that the constituent gates, from which they inherit this error, are optimised to have this level of phase acquisition error.

To examine when this crosstalk develops, Fig. 4.6 compares the ion displacements with the accumulated pair phases for supersonic and subsonic implementations. The accumulated phases $\Theta_{j\ell}$ are calculated using Eq. (3.17), including all SDKs applied up to each time t. In the interaction picture with respect to the free motional evolution, the accumulated phases change only when an SDK is applied, giving vertical steps at SDK events and horizontal segments during free evolution. In the supersonic gate, motion remains highly localised around the addressed pairs, with the spectator ions separating the pairs experiencing negligible displacement. The intended entangling phases reach their target values while crosstalk remains close to zero throughout the sequence. This

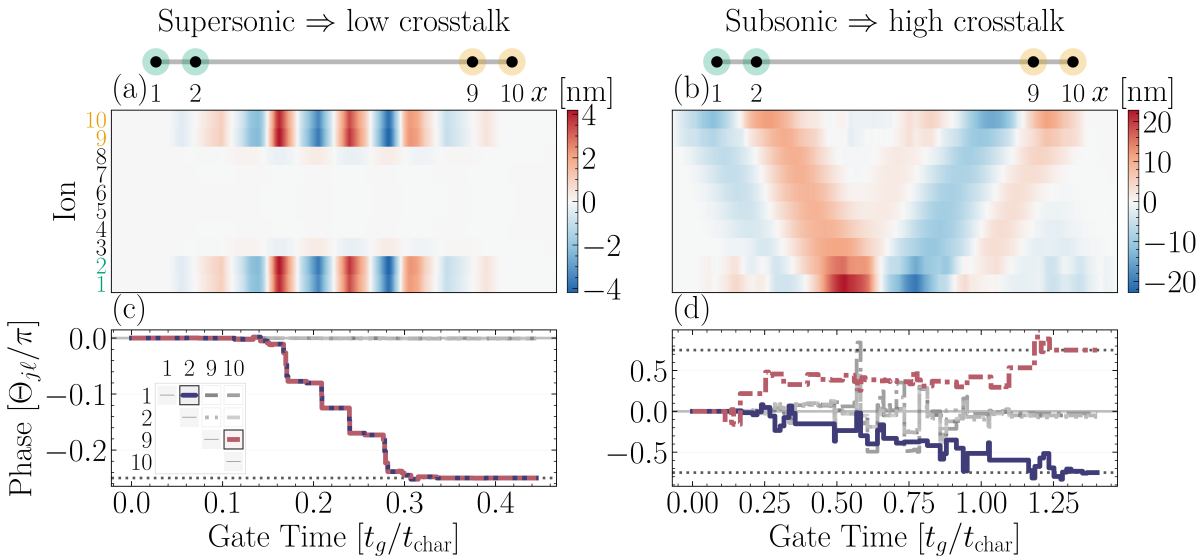


Fig. 4.6: Naive parallelisation of well-separated ion pairs $(1, 2)$ and $(9, 10)$ in the supersonic (left-hand panels) and subsonic (right-hand panels) regimes. Signed ion displacements (note the different colour scales), in the supersonic (a) and subsonic (b) regime. As in Fig. 2.7, supersonic motion remains localised with weak crosstalk, whereas subsonic motion is highly non-localised with crosstalk developing at around $t_{\text{char}}/2$. (c) and (d) present the accumulated entangling phases $\Theta_{j\ell}/\pi$ for target and crosstalk interactions. Horizontal dotted lines indicate maximally entangling phases, $\Theta = \pi/4 \pmod{\pi/2}$.

is consistent with the gate ending before appreciable motion spreads between the pairs, resulting in a small final phase error.

In the subsonic gate, motion extends throughout the chain and substantial crosstalk develops during the sequence, particularly around $t_{\text{char}}/2$. As t_{char} represents approximately the round-trip propagation time along the chain, $t_{\text{char}}/2$ corresponds to the time required for a disturbance generated near one end to reach the opposite end of the ion chain. The appearance of crosstalk around this time is therefore consistent with motion generated at one addressed pair reaching the other. Furthermore, crosstalk evolves irregularly, with contributions from successive SDKs partially cancelling. However, this cancellation is incomplete, leaving residual crosstalk that limits the final gate fidelity. The correspondence between the spreading motion and the accumulation of crosstalk therefore supports the interpretation that shared motional dynamics mediate crosstalk.

Although these sequences were optimised for the individual gates, the partial cancellation of crosstalk suggests that their final values may depend sensitively on the relative timing of the two pulse sequences. Indeed, introducing a relative delay Δt between the

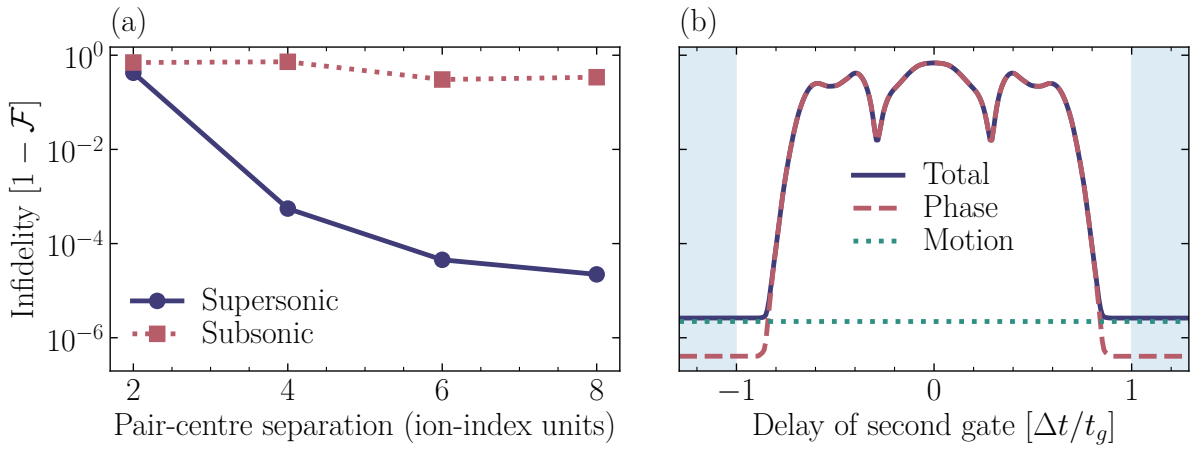


Fig. 4.7: Dependence of two-gate infidelity on spatial separation and relative timing. (a) Total infidelity for simultaneous supersonic and subsonic gates as a function of pair-centre separation for local pairs. Here, each data point corresponds to a gate from different geometry tested. In particular, $(1,2)$ and $(9,10)$, $(2,3)$ and $(8,9)$, $(3,4)$ and $(7,8)$, $(4,5)$ and $(6,7)$ are the geometries used. Gate infidelity for supersonic gates is highly dependent on the separation between ion pairs. Subsonic gates are invariant to the separation as they necessarily require motion to be non-localised regardless of pair-separation or geometry. (b) Total, phase and motional infidelities as the second gate is delayed by Δt, with the delay normalised by t_g. The phase infidelity varies irregularly before becoming comparable to the constant motional infidelity. Here, gates that entangle ions $(1,2)$ and $(9,10)$ are used.

sequences changes their crosstalk accumulation without altering the pulse timings within either individual gate. This motivates investigating whether naive parallelisation can be improved by increasing the separation between the addressed pairs or introducing a free relative timing delay, without requiring a costly numerical optimisation approach that is designed to suppress crosstalk.

4.4 Spatial separation and relative timing

The preceding analysis links the failure of naive parallelisation to crosstalk generated through shared motional dynamics. I now examine whether spatial separation and relative timing can reduce these errors without re-optimising the constituent pulse sequences.

4.4.1 Spatial separation

The left panel of Fig. 4.7 shows the infidelity of simultaneous local gates as a function of pair-centre separation, measured in ion-index units. For the supersonic gates considered here, increasing the separation reduces the infidelity by approximately five orders of magnitude, from order unity for adjacent pairs to approximately 10^{-5} for the most widely separated pairs. By contrast, the subsonic gates retain comparatively large infidelities for the same target pairs. This supports the interpretation that spatial separation is effective when the driven motion remains sufficiently localised over the gate duration. For subsonic gates, however, the motion necessarily extends throughout the chain, as found in Fig. 4.6, providing a pathway for crosstalk even between widely separated pairs.

4.4.2 Relative timing

Thus far, naive parallelisation has involved starting both gates simultaneously. However, introducing a relative delay Δt provides a single free parameter that can be varied to minimise crosstalk without altering the constituent pulse sequences or requiring a computational expensive optimisation. This is implemented by translating every pulse time of the second gate according to

$$t_m^{(k\ell)} \longrightarrow t_m^{(k\ell)} + \Delta t. \tag{4.3}$$

The shift preserves all SDK strengths and timings within the second gate. Therefore this preserves the motional-restoration and target-phase conditions of each constituent gate.

The right panel of Fig. 4.7 shows that the phase infidelity varies non-monotonically with the relative delay, showing improvements in total infidelity from order unity to 10^{-2} but only falls below the approximately constant motional contribution when the gates no longer overlap. For two gates of duration t_g, this occurs when $|\Delta t| \gtrsim t_g$, corresponding to sequential execution (i.e. the gate are no longer in parallel). In this case, crosstalk vanishes for perfectly motionally restored constituent gates, leaving only their individual errors. Although varying Δt can improve the infidelity, a single timing parameter is clearly insufficient to simultaneously suppress all four crosstalk terms when $|\Delta t| < t_g$ (i.e.

for parallel operation).

Together, these results show that spatial separation can enable high-fidelity supersonic parallelisation. The introduction of a relative delay can provide improvements in performance for the gates considered here but does not recover high-fidelity parallel operation. This motivates the dedicated optimisation developed in Chapter 5, where the pulse sequences themselves are adjusted to suppress crosstalk while preserving the intended entangling operations.

4.5 Conclusions

In this chapter, I tested a naive approach to parallelising two-qubit fast gates by applying existing high-fidelity two-qubit solutions simultaneously and evaluating the result using the multi-qubit fidelity developed in Chapter 3. Of the four representative ion-pair geometries considered, only well-separated local pairs achieved low infidelity but only in the supersonic regime. Phase errors dominated the poor-performing gates at longer times, while their motional error remained comparatively small.

The well-separated pairs revealed the source of this phase error. Their desired entangling phases remain accurate ($|\Delta\Theta_{j\ell}| < 10^{-4}$) in both regimes, but the crosstalk is roughly two orders of magnitude larger in the subsonic regime. During this gate, crosstalk develops before t_{char} and partially cancels by the end of the sequence. The accompanying ion motion spreads through the chain, consistent with shared motional modes mediating crosstalk.

The spatial separation and relative timing delay further clarify the conditions governing naive parallelisation. Increasing the separation between the addressed pairs strongly reduces the supersonic infidelity, while the subsonic infidelity remains comparatively large across the separations considered. Delaying the second gate changes the phase contribution non-monotonically before reducing it below the approximately constant motional contribution once Δt exceeds the gate time of the first gate, i.e. the gates are no longer applied in parallel but sequentially.

These results show that the performance of independently optimised gates depends on both their spatial arrangement and their relative timing. The partial cancellation

of crosstalk also motivates a search for pulse sequences that suppress crosstalk explicitly while retaining overlapping operation. In the following chapter, I optimise pulse sequences using the full parallel-gate cost function to determine whether crosstalk can be suppressed in the regimes where naive parallelisation fails.

Chapter 5

Parallel fast gates for arbitrary ion pairs

In Chapter 4, I investigated whether existing high-fidelity two-qubit fast gates could be applied simultaneously without reoptimisation to achieve parallel operation. Under the pulse constraints considered, well-separated local pairs achieved low parallel infidelity in the supersonic regime, while adjacent and non-local geometries failed to achieve comparable infidelities at both subsonic and supersonic timescales. I showed that crosstalk errors dominated which were the reason for decreased fidelities. Crosstalk developed only in the subsonic regime where motion necessarily spreads through the chain. It is therefore the main obstacle to parallel operation of non-local fast gates.

Suppressing such crosstalk is a central constraint in previous work on parallel adiabatic trapped-ion entangling gates [1, 77]. As shown in Chapter 4, crosstalk evolution in the impulsive fast-gate schemes is essentially random when two independently optimised two-qubit gates are simultaneously applied. It is therefore necessary to extend the gate search protocol (recall § 2.6.4) to optimise gate schemes for parallel operation using the multi-qubit infidelity developed in Chapter 3 as the cost function. The goal is to accumulate the desired entangling phases, restore the motion, and suppress crosstalk while remaining within experimentally motivated SDK constraints. This allows me to identify whether numerical optimisation can produce high-fidelity parallel gates in geometries where naive parallelisation fails, and whether these gates offer an advantage over their serial counterparts.

This chapter is organised as follows. Section 5.1 describes the parallel search protocol and its pulse constraints. In § 5.2, I compare these optimised parallel gates with the naive

parallelisation protocol from Chapter 4 for representative pair geometries. Section 5.3 then compares the gate times and SDK requirements of optimised parallel gates with the equivalent serial implementations of the same two-qubit gates. Finally, § 5.4 summarises the feasibility and resource requirements of optimised parallel fast gates.

5.1 Parallel fast-gate search protocol

I developed an independent implementation of the 'fast-entangling ion gate and robustness optimiser' algorithm which is explained in § 2.6.4 and used in previous fast-gate investigations [29, 65–67] and extended it to optimise parallel gates. In the two-qubit formulation, a gate scheme is represented by SDK strengths and times, $\{z_k, t_k\}$. My implementation extends this representation to $\left\{z_k^{(p)}, t_k^{(p)}\right\}_{p\in\mathcal{P}}$, where $\mathcal{P}$ is the set of M target ion pairs. A separate SDK sequence is assigned to each target pair, with all sequences optimised by minimising the parallel-gate infidelity derived in Chapter 3 (see Eq. 3.33),

$$J\left(\{z_k^{(p)}\}, \{t_k^{(p)}\}\right) = 1 - \mathcal{F}\left(\{z_k^{(p)}\}, \{t_k^{(p)}\}\right), \tag{5.1}$$

Recall that, this is the same fidelity measure used to evaluate the gates produced by the naive parallelisation protocol. It therefore penalises residual motion, deviations from the desired entangling phases, and critically, crosstalk.

As illustrated in Fig. 4.6, crosstalk can evolve substantially during a gate yet partially cancel by its completion. This suggests that appropriately adjusting the SDK sequences should suppress residual crosstalk while maintaining the desired entangling phases.

The searches in § 5.2 use the same per-pair pulse constraints as imposed for the naively parallel gates: 48 pulse groups per target pair, with at most 5 SDKs per group. The same two-stage optimisation process as previous work is employed (recall Fig. 2.6). The implementation assumes independent control of each pair's SDK sequence, with individual sequence durations constrained to lie within 20% of a common target gate duration to facilitate parallel execution. Imposing the same pulse constraints as those in Chapter 4 allows a direct comparison with the naive parallelisation protocol, isolating the effect of explicitly suppressing crosstalk during optimisation. The restriction on the

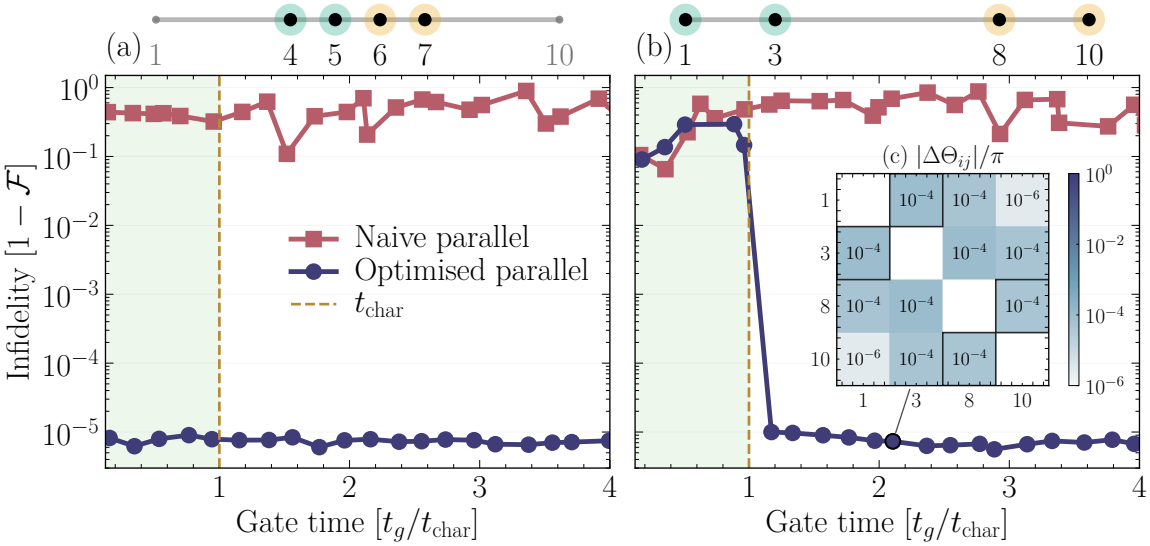


Fig. 5.1: State-averaged infidelity for (a) adjacent local pairs $(4, 5)$ and $(6, 7)$ and (b) non-local pairs $(1, 3)$ and $(8, 10)$ in a ten-ion chain. Red squares show naive parallelisation and blue circles show numerically optimised parallel gates. Dashed vertical lines mark $t_g = t_{\text{char}}$, with the supersonic regime shaded green. Optimised results cluster near the 10^{-5} stopping threshold for local pairs in both regimes and non-local pairs in the subsonic regime. (c) Deviations from the $|\Delta\Theta_{ij}|/\pi$ for the highlighted subsonic gate at $t_g = 6.99\,\mu\text{s}$, with logarithmic colouring and annotations rounded to orders of magnitude. Target-pair entries show intended-phase errors; other off-diagonal entries show crosstalk.

number of SDKs is also experimentally motivated, as increasing the pulse count can improve idealised gate fidelity at the expense of accumulating errors from imperfect SDK operations which have limited SDK demonstrations [50].

5.2 Existence of high-fidelity parallel gate solutions

Fig. 5.1 compares naive parallelisation with the optimisation protocol outlined in § 5.1, applied to adjacent local pairs $(4, 5)$ and $(6, 7)$ and non-local pairs $(1, 3)$ and $(8, 10)$. Solutions obtained using this protocol are hereafter referred to as 'optimised'. Optimisation terminates when the truncated infidelity reaches 10^{-5}, explaining the clustering of successful solutions near this value. Lower targets were not considered because corrections neglected by Coulomb linearisation (recall § 2.3) can affect fast-gate infidelities at order 10^{-6} [65], requiring further optimisation with the full Coulomb interaction to assess

lower infidelities.

For adjacent local pairs, numerical optimisation reduces the parallel infidelity by approximately five orders of magnitude, achieving the infidelity target across the entire range of gate times considered. In contrast, non-local pairs exhibit a sharp improvement near t_{char}, with high-fidelity solutions found only for subsonic gates. As introduced in § 2.7, this behaviour is consistent with the time required for collective motion to mediate the intended non-local interactions and subsequently restore the motion. Importantly, Fig. 5.1 evidences that it is necessary for motion to spread throughout the ion chain for crosstalk suppression to occur for non-local gates. Moreover, as non-local gates with an experimentally feasible amount of SDKs are not possible in the supersonic regime, the high-infidelities are not a consequence of crosstalk but of insufficient time to acquire the desired entangling phases [29].

Fig. 5.1(c) illustrates this suppression for an optimised non-local gate at $t_g = 6.99\,\mu\text{s} > t_{\text{char}}$, where both target-phase errors and crosstalk are of order $10^{-4}\pi$ or smaller. Together with the low total infidelity, this demonstrates that numerical optimisation can suppress crosstalk while satisfying the phase and motional restoration conditions required for high-fidelity parallel gates.

Additional searches, not shown, found high-fidelity solutions for other non-local pair geometries with similar qualitative gate-time dependence. These results demonstrate the existence of high-fidelity parallel solutions for several representative geometries within the imposed pulse constraints. Section 5.3 examines whether these solutions also offer a shorter gate duration or reduction in SDK requirements over serial execution.

5.3 Benchmarking parallel against serial gates

Having established the existence of high-fidelity parallel gates in § 5.2, I next investigate whether they offer a resource advantage over serial implementations of the same two-qubit fast gates. Although naive parallelisation can achieve high fidelities for some pair geometries, it does not do so across both gate time regimes. The relevant comparison for numerically optimised parallel gates is therefore the serial application of the two constituent gates. The important resources are the total gate time and the total number

of SDKs, both of which should be minimised. Shorter gate times are desirable as this increases the absolute number of gates that can be performed within the coherence time of the ions. Minimising the number of SDKs is also essential in ensuring high-fidelity gate operation as each SDK has an associated error which compounds with each SDK.

Parallelisation provides a timing advantage when both gates can be completed faster than their serial counterparts, with a twofold speedup providing a natural benchmark for two equal-duration gates. In § 5.3.1, I explain the algorithm implemented to identify solutions with favourable trade-offs between SDK count and gate time. The resulting resource requirements are compared in § 5.3.2 to assess the advantages of parallel over serial implementation.

5.3.1 Protocol for gate resource analysis

The parallel fast-gate schemes presented in § 5.2 were obtained by minimising Eq. (5.1) to constraints on the number of SDKs. As imperfections in individual SDKs can compound with the number of pulses [50, 65], I use the total number of SDKs as a proxy for pulse-error rather than explicitly introducing these errors into a more complicated cost function. By fixing a fidelity threshold motivated by the requirements of fault-tolerant quantum computation [38, 40], I investigate the trade-off between gate duration and SDK count. This is achieved by constraining the number of SDKs and searching for the shortest gate duration that satisfies the prescribed fidelity threshold. I developed an independent implementation of a recently introduced search procedure from the ANU fast-gate theory group, summarised in Fig. 5.2.

For each search, I fix the number of pulse groups per target pair and impose an integer upper bound B on the SDK strength of each group $z_k^{(p)}$, such that $|z_k^{(p)}| \leq B$. The total SDK count of a parallel gate is given by $\mathcal{N} = \sum_{p\in\mathcal{P}} \sum_k |z_k^{(p)}|$, so this bound constrains the maximum number of SDKs without fixing the number actually used. Increasing B therefore enlarges the allowed search space without necessarily increasing the SDK count of the resulting gate schemes.

I search over gate durations from 2 µs to 15 µs, encompassing both the supersonic and subsonic regimes. Within this window, I sample evenly spaced gate times and numerically

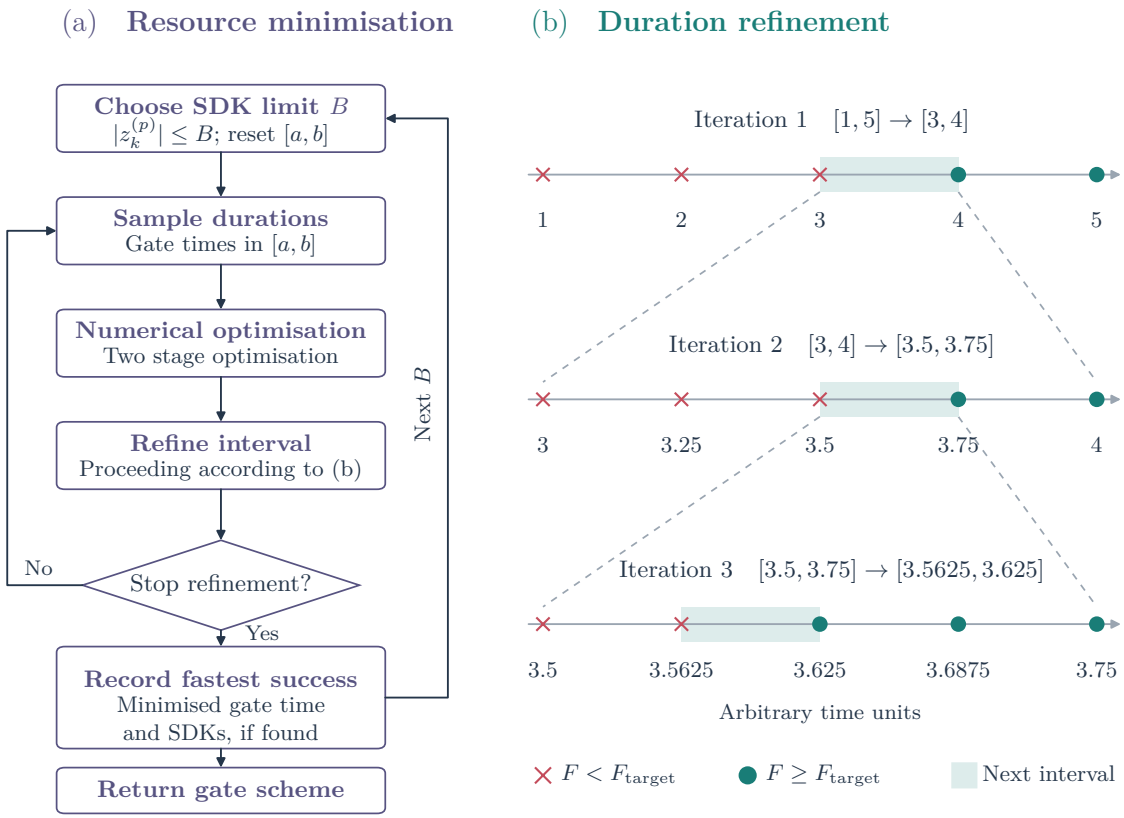


Fig. 5.2: Resource minimisation search over gate time and SDK budget. (a) For each limit B, gate sequences are optimised from random initialisations subject to $|z_k^{(p)}| \leq B$. The duration interval is refined between the shortest gate time that satisfies the fidelity threshold and the longest gate time that fails to meet the threshold. This is performed until the time tolerance or iteration limit is reached, then reset for the next budget. (b) Illustrative refinement with five samples per iteration. Crosses denote failures, circles successes, shading indicates the next interval, and dashed lines the next refinement gate time interval. Durations use arbitrary units. As with all gate searches in this thesis, global optimality is not established.

optimise the gate sequences from random initialisations, subject to the pulse-group and SDK-strength constraints. A solution is classified as successful if it reaches the prescribed fidelity threshold, with the specific thresholds considered in § 5.3.2.

The search interval is then refined between the shortest successful gate and the nearest shorter tested duration that failed to reach the fidelity target. If no successful solution is found, subsequent samples are concentrated in the longer-duration half of the current interval. Conversely, if a successful solution is found without a shorter unsuccessful result, refinement continues between the original lower time limit and the shortest successful duration. Refinement terminates when the interval reaches the specified time tolerance

or the maximum number of iterations is exhausted. The procedure is then repeated from the original time window with B increased by one.

I apply this procedure to both parallel and serial implementations. For the serial implementation, the two target pairs are optimised separately according to this procedure, with candidate sequences combined to give total duration and SDK count of $t_{\text{serial}} = \sum_{p\in\mathcal{P}} t_p$ and $\mathcal{N}_{\text{serial}} = \sum_{p\in\mathcal{P}} \mathcal{N}_p$, respectively. Both implementations must satisfy the same fidelity target for the complete two-gate operation. The reported resource requirements represent the best solutions identified by the numerical searches and do not establish global minima.

5.3.2 Resource savings from parallel fast gates

I apply the method described in § 5.3.1 to two representative geometries, comprising well-separated local pairs and non-local pairs. These cases allow me to assess resource savings both where naive parallelisation was previously effective and where optimisation is required to suppress crosstalk. I compare optimised parallel gates with the serial application of separately optimised two-qubit gates. This provides a baseline for assessing the resource advantages of parallel operation, while also allowing a comparison with naive parallelisation where high-fidelity solutions are available.

The searches consider gate durations from 2 µs to 15 µs, with 48 pulse groups per pair and SDK strengths incremented from $B = 1$ to $B = 4$, corresponding to an upper limit of $48 \times 4 = 192$ SDKs per target pair. For serial implementations, the total gate time and SDK count are summed over the two constituent gates. I consider fidelity thresholds of 90%, 99%, and 99.9% for both serial and parallel gates between non-local pairs. In the local-pair case, there is little fidelity dependence, so only 99% and 99.9% are considered. The higher thresholds of 99% and 99.9% are motivated by the accuracy requirements of fault-tolerant quantum computation [38, 40]. These values are used as physically motivated performance targets, without assuming that the fidelity of the complete operation alone establishes compatibility with a particular error-correcting code. The 90% fidelity threshold provides a lower-fidelity reference for assessing how resource requirements change as the accuracy requirement is relaxed, while remaining well be-

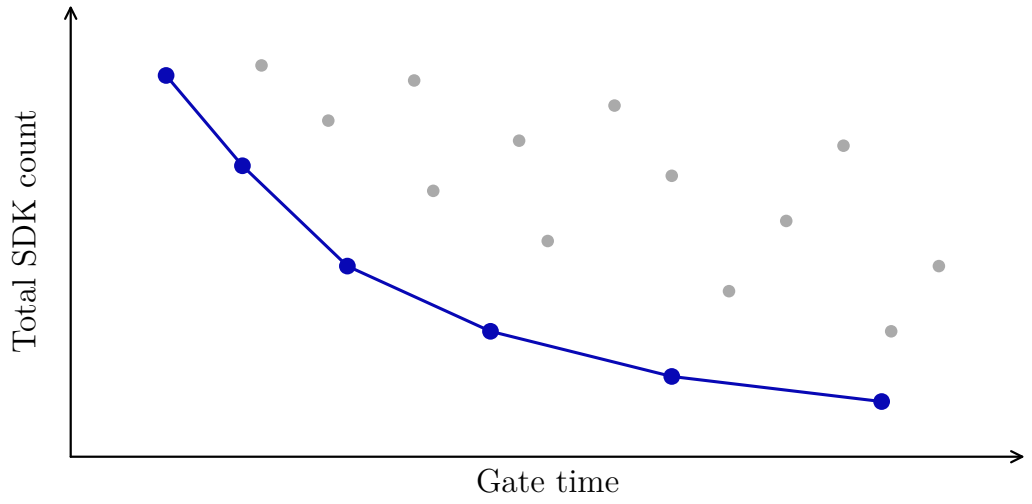


Fig. 5.3: Illustration of the construction of a Pareto front for two desirable quantities. Blue points represent solutions for which neither cost can be reduced without increasing the other among the sampled solutions. Grey points are dominated by at least one candidate with equal or lower values of both costs and a strictly lower value of at least one. The connecting line guides the eye and does not imply that intermediate solutions are available. In the resource analysis outlined in § 5.3.1 and performed in § 5.3.2, the two costs are gate time, t_g, and total SDK count, $\mathcal{N}$, at a specified minimum fidelity.

low the gate fidelities typically required by practical fault-tolerant quantum computing schemes [78].

For each fidelity threshold, I construct Pareto fronts (see Fig. 5.3) in gate time, t_g, and total SDK count, $\mathcal{N}$. These retain the successful solutions for which neither resource can be reduced without increasing the other among the candidates found. Comparing the serial and parallel fronts reveals whether parallel operation reduces the SDK count at a fixed completion time, shortens the operation at a fixed SDK budget, or introduces a trade-off between the two resources.

Well-separated local pairs

I first consider the well-separated local pairs $(1, 2)$ and $(9, 10)$. Fig. 5.4 compares serial execution, numerically optimised parallel gates, and naive parallelisation at fidelity thresholds of 99% and 99.9%. The naive parallelisation protocol is included because, for this geometry, I showed in Chapter 4 that high-fidelity parallel gates exist in the

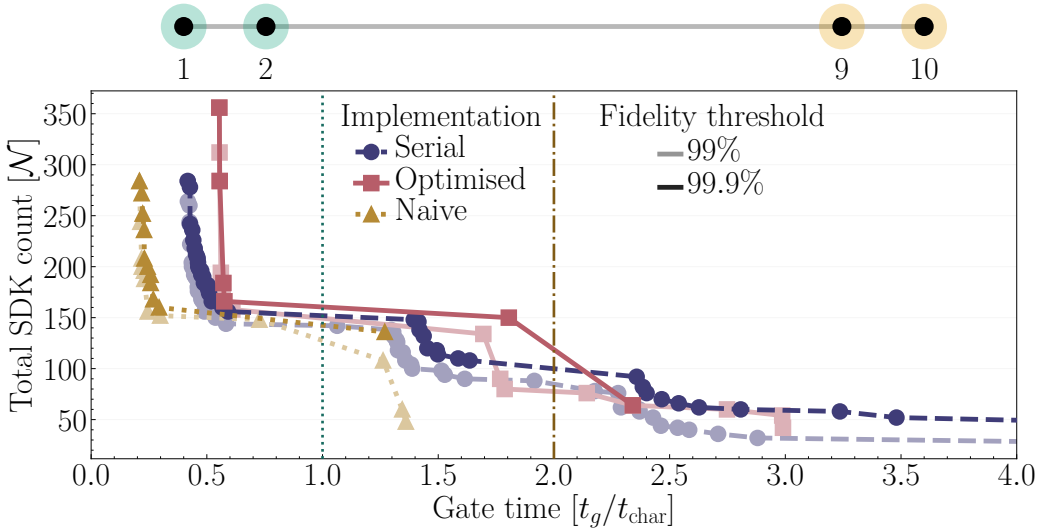


Fig. 5.4: Gate time and SDK requirements for well-separated local pairs $(1, 2)$ and $(9, 10)$ in a ten-ion chain. Pareto fronts compare serial execution (circles), numerically optimised parallel gates (squares), and naive parallelisation (triangles), at minimum fidelities of 99% (lighter) and 99.9% (darker). Markers indicate retained candidates noting that connecting lines guide the eye. Serial durations and SDK counts are summed over both constituent gates. Vertical lines mark t_{char} and $2t_{\text{char}}$, where t_{char} is the characteristic sound-propagation time of the chain.

supersonic regime.

Naive parallelisation achieves the shortest durations shown and offers favourable SDK requirements, demonstrating that reusing gates optimised for two-qubit entanglement can provide a resource advantage for this well-separated geometry. This is because the ion motion within the pairs is highly localised meaning that for these well-separated local pairs the gates effectively operate independently without influencing each other. Interestingly, in contrast, numerical optimisation does not produce a consistent improvement over serial execution, as serial candidates generally achieve comparable or lower SDK counts at a given completion time, although the Pareto fronts cross in some regions. The reference times t_{char} and $2t_{\text{char}}$ do not impose lower bounds on the achievable gate durations. This is consistent with local gates operating in the supersonic regime, where entanglement between neighbouring ions does not require motion to propagate across the entire chain.

The comparatively poor performance of optimised parallel gates reflects limitations of the numerical search. Since naive parallelisation provides feasible solutions within the same search space, its superior resource efficiency suggests that the parallel optimiser

struggles to identify comparably efficient solutions. Although all searches used 384 random initialisations, this does not ensure comparable coverage of the optimisation spaces, particularly given the additional crosstalk-cancellation conditions that must be satisfied. Furthermore, requiring the pulse sequences to start simultaneously in the optimised parallel gates restricts their relative timing and may exclude more resource-efficient solutions. These results therefore demonstrate the resource efficiency of naive parallelisation for well-separated local pairs, while suggesting that more extensive searches or greater flexibility in pulse timing may improve the optimised parallel solutions. These possibilities are discussed further in Chapter 6.

Non-local pairs

The non-local pairs $(1, 4)$ and $(7, 10)$ in Fig. 5.5 reveal a fidelity-dependent timing advantage for parallel operation. The shortest serial solutions occur near $2t_{\text{char}}$, consistent with each constituent non-local gate requiring approximately t_{char} or longer among the solutions found. By contrast, the shortest parallel solutions reaching 90%, 99%, and 99.9% fidelity occur near t_{char}, $1.3t_{\text{char}}$, and $2t_{\text{char}}$, respectively. Parallelisation therefore offers a substantial timing advantage at lower fidelity thresholds, approaching a twofold speedup, but this advantage diminishes as the required fidelity increases.

The heatmaps in the bottom panels of Fig. 5.5 further illustrate this dependence across SDK budgets. The boundary of successful serial solutions changes comparatively little with fidelity, whereas more stringent parallel requirements shift this boundary towards longer durations. The fastest parallel solutions also require relatively large SDK counts, increasing their sensitivity to pulse imperfections. Longer durations reduce these requirements, but also allow serial execution to access lower-resource solutions. This suggests that simultaneously satisfying the phase, motional restoration, and crosstalk suppression conditions becomes increasingly demanding for high-fidelity parallel gates at short gate times.

Fig. 5.6 further examines the dependence of gate duration on fidelity at total SDK budgets of 100 and 300. For each infidelity threshold, t_{min} is the shortest qualifying duration found within the specified budget, with both serial and parallel durations measured relative to t_{char}. At higher fidelity thresholds, increasing the SDK budget from 100 to

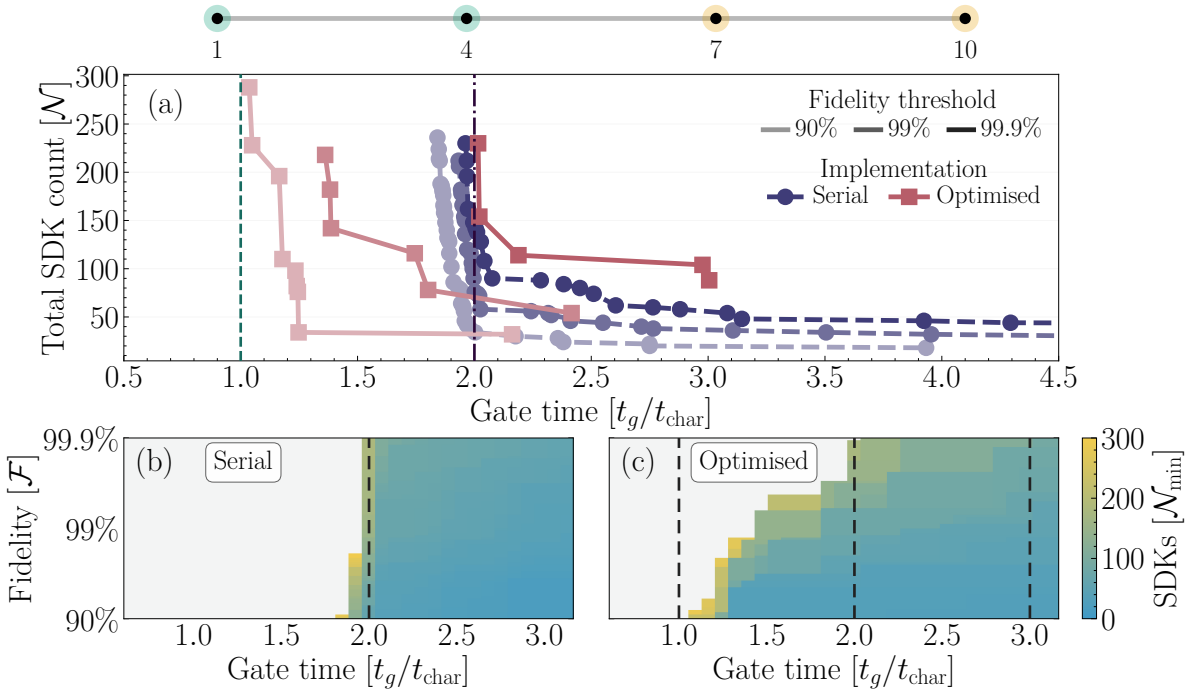


Fig. 5.5: Gate time and SDK requirements for non-local pairs $(1, 4)$ and $(7, 10)$ in a ten-ion chain. (a) Pareto fronts for serial (circles) and numerically optimised parallel (squares) execution at minimum fidelities of 90%, 99%, and 99.9%, indicated by increasingly dark shades. Serial durations and SDK counts are summed over both gates. The bottom panels show the minimum total SDK count found as a function of gate time and fidelity threshold for serial (b) and parallel (c) execution, using a shared colour scale. Uncoloured regions indicate that no qualifying candidate was found within the search constraints. Vertical reference lines mark integer multiples of t_{char}.

300 reduces the parallel duration offset from approximately $2t_{\mathrm{char}}$ to t_{char}. Relaxing the fidelity requirement permits shorter parallel sequences, approaching t_{char}, whereas serial solutions remain comparatively close to $2t_{\mathrm{char}}$. Thus, the timing advantage of parallelisation depends on both fidelity and SDK budget, diminishing as more stringent fidelity requirements are imposed. The step-like changes reflect transitions between the discrete candidates found by the search rather than discrete allowed gate durations.

Unlike serial gates, the parallel solutions exhibit changes in their minimum achievable durations near multiples of t_{char}. This may reflect the additional phase constraints imposed by parallel operation, where the desired entangling phases must be established while simultaneously suppressing crosstalk. Longer gate durations may provide greater

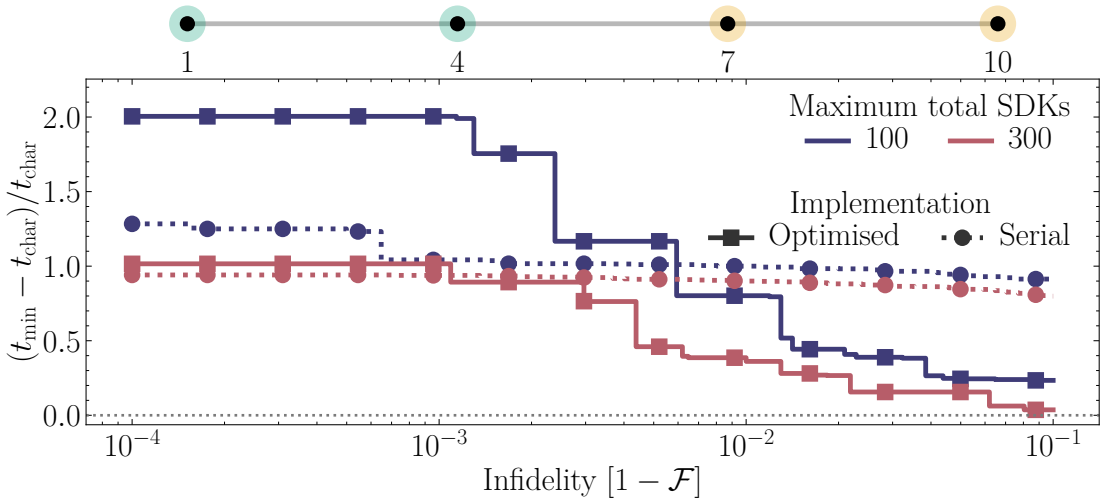


Fig. 5.6: Minimum gate time ($t_{\min}$) with respect to the characteristic time (t_{char}), $(t_{\min} - t_{\text{char}})/t_{\text{char}}$ required to satisfy a given infidelity threshold for non-local gates on $(1, 4)$ and $(7, 10)$, identified in the overhead schematic. Squares denote parallel implementations and circles denote serial implementations. Colours indicate maximum total SDK counts of 100 and 300. Larger SDK budgets generally reduce the gate time required to meet the fidelity target but come at the cost of an increase in the associated SDK error.

opportunities for the driven motional dynamics to satisfy these competing conditions, potentially explaining the observed dependence on t_{char}. These additional constraints also make the optimisation problem more challenging, particularly at high fidelity thresholds. However, the apparent transitions may also reflect the finite density of the numerical searches, with more favourable solutions potentially remaining undiscovered. Further investigation of the phase evolution and motional trajectories, alongside more extensive numerical searches, would be needed to establish whether these features arise from underlying physical dynamics or numerical limitations.

5.4 Conclusions

This chapter establishes existence of high-fidelity parallel fast gates and assesses their resource requirements relative to serial execution. Numerical optimisation produces solutions meeting the prescribed fidelity target in both gate-time regimes for local pairs and at the sampled subsonic durations for non-local pairs. These results demonstrate that

crosstalk can be suppressed while acquiring the desired entangling phases and restoring the motion.

The existence of high-fidelity solutions does not, however, guarantee a resource advantage. Comparing the Pareto fronts in total SDK count and gate time reveals that numerically optimised parallel gates do not consistently outperform serial execution for the well-separated local geometry studied, although naive parallelisation provides favourable resource requirements in the supersonic regime. For the non-local geometry, optimised parallel solutions offer an advantage at selected fidelity thresholds and gate times, particularly within $t_{\mathrm{char}} < t_g < 2t_{\mathrm{char}}$, where no serial solutions were found. This advantage diminishes as the fidelity requirement increases and the successful parallel sequences require longer durations and larger SDK budgets. The benefit of parallelisation therefore depends on the target geometry, required fidelity, and available pulse resources.

Chapter 6

Discussion and conclusions

Of the many quantum computing platforms that have been proposed, trapped ions are among the most mature, having demonstrated high-fidelity two-qubit entangling gates and high qubit connectivity. A requirement for implementing scalable quantum computation is going beyond sequentially applying gates and instead performing many gates in parallel, thereby reducing circuit depth and increasing the overall amount of computations within the finite qubit coherence time. This thesis has demonstrated both the existence and resource requirements of parallel high-fidelity fast two-qubit entangling gates, representing a meaningful step towards their implementation in near-term trapped-ion quantum processors.

6.1 Key findings and physical implications

The results of this thesis demonstrate that the feasibility of naive parallelisation is strongly governed by the spatial extent of collective motional dynamics. The shared motional modes that mediate entanglement between arbitrary ions also provide a pathway for crosstalk which degrades the fidelity of parallel operations. However, for well-separated nearest-neighbour pairs operating in the supersonic regime, motional disturbances remain largely localised around the addressed ions. Consequently, the parallel gate fidelities are comparable to those obtained through serial execution, demonstrating that independently optimised well-separated fast gates can be implemented simultaneously without substantial crosstalk-induced errors. In contrast, naive parallelisation of non-local gates fails to achieve high fidelity across the investigated gate-time regimes. In the supersonic regime, non-local gates cannot accumulate the required entangling phases with experimentally feasible SDK resources, while in the subsonic regime, the propaga-

tion of collective motion throughout the ion chain introduces substantial crosstalk. This highlights an important trade-off in that the collective phonon propagation that enables high-connectivity fast gates can also inhibit their parallel execution.

The limitations of naive parallelisation can be overcome through numerical optimisation of the pulse sequences, explicitly accounting for crosstalk. Using the multi-qubit theoretical framework developed in Chapter 3, I identify high-fidelity parallel fast-gate solutions across the investigated ion-pair geometries, including configurations for which naively parallelised gates exhibit significant crosstalk. This demonstrates that the failure of naive parallelisation does not imply a fundamental inability to implement simultaneous entangling operations. Rather, the additional crosstalk interactions must be incorporated into the optimisation procedure. However, satisfying these additional constraints introduces a more demanding optimisation problem and can increase the pulse resources or gate durations required to achieve high fidelity. In particular, parallel gates involving non-local pairs remain strongly influenced by the characteristic phonon propagation timescale, with the fastest identified solutions occurring on the order of t_{char}.

Although numerically optimised parallel gates can suppress crosstalk, their practical advantage depends on the trade-off between gate duration, fidelity and SDK resources. For example, parallel operations of non-local pairs achieving 99% fidelity were identified with durations as short as $1.3t_{\mathrm{char}}$ using 140 SDKs, compared with approximately $2t_{\mathrm{char}}$ for the equivalent sequential implementation. This demonstrates that optimised parallelisation can provide a meaningful reduction in execution time. However, increasing the fidelity requirement substantially reduces this advantage. At a fidelity threshold of 99.9%, the shortest identified non-local parallel solutions require durations comparable to sequential execution under the imposed pulse constraints. Furthermore, the non-convex optimisation landscape and undersampling of parallel solutions, particularly for well-separated local pairs, mean that these results do not establish a fundamental limit on the achievable parallel speed-up. Nevertheless, they indicate that the benefits of parallelisation become increasingly difficult to realise as more stringent fidelity requirements are imposed.

6.2 Limitations and future directions

This thesis presents a theoretical demonstration of parallel fast entangling gates and the necessary SDK resources required for their implementation. It does not, however, establish the parallel fast gates studied as an inherently superior means to quantum information processing using trapped ions due to several limitations which I now outline. I also discuss how one can take these limitations and turn them into avenues for future work which would further assess the feasibility of parallel fast entangling gates.

Computational scaling of parallel fast gates

Throughout this thesis, I consider chains of 10 ions and restrict the main investigation to at most two simultaneous two-qubit gates. This system size provides a compromise between the increasing control complexity associated with longer ion chains and the ability to investigate different spatial configurations of parallel gates. Furthermore, studying two simultaneous gates is sufficient to isolate the fundamental mechanisms of crosstalk, identified in Chapters 3 and 4 as a principal error source. This is consistent with Figgatt *et al.* [1] where the suppression of crosstalk proved experimentally challenging. Only considering two simultaneous gates leaves much of the potential parallelism of the chain unexplored, as 10 ions can support up to five disjoint two-qubit gates simultaneously. Extending the investigation to this limit would help establish how gate fidelity and control requirements scale with the degree of parallelisation. A significant challenge, however, is the increasing computational cost of identifying suitable pulse sequences. As shown in Fig. 6.1 as a preliminary result, the mean optimisation cost increases rapidly with the number of simultaneous gates, even at the relatively low fidelity threshold of 90%. Although these results do not establish the asymptotic computational scaling, they suggest that more efficient optimisation methods may be necessary to realise the full parallelisation potential for fast gates in larger trapped-ion systems. Indeed, even in the case of two well-separated pairs in parallel the results in § 5.3.2 demonstrated that the current optimisation protocol is unable to identify global minima and severely undersamples the search space as all optimised solutions were outperformed by even the naive parallelisation protocol. Whilst initialising more random guesses is a method for increasing the

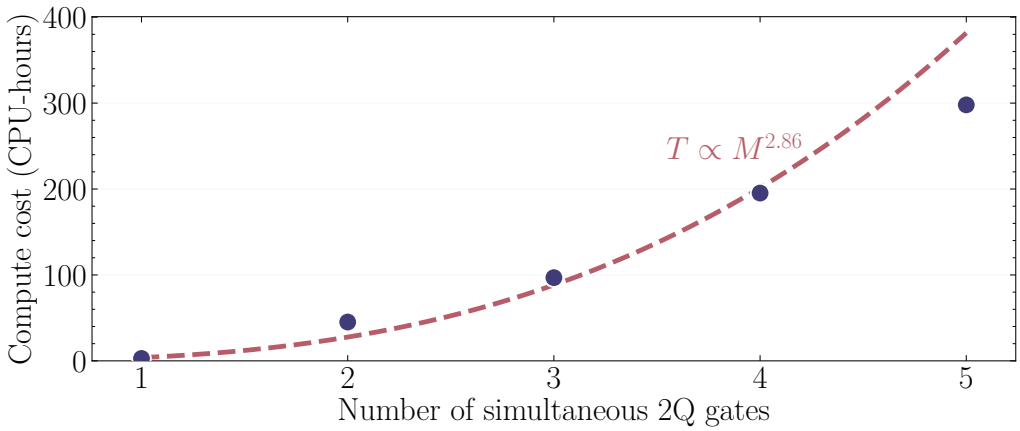


Fig. 6.1: Mean computational cost of numerically optimising fast-gate sequences as a function of the number of simultaneous two-qubit gates, M. Computational cost is reported in CPU-hours. All searches were performed using 384 CPU cores with a minimum fidelity threshold of 90%. The dashed curve illustrates an approximate empirical power-law dependence, $T \propto M^{2.86}$, over the investigated range and does not represent a universal scaling law.

search density, this will not scale well with many pairs in parallel emphasising the need to improve the optimisation protocol for these more computationally intensive searches.

An avenue for addressing the limited search density is to provide the optimiser with independently optimised, high-fidelity two-qubit gates as initial guesses before applying the parallel gate optimisation procedure. This is motivated by the observation that well-separated local ion pairs already perform well under the naive parallelisation protocol. Furthermore, I showed that adjacent local pairs, which suffer substantial crosstalk under naive parallelisation, can nevertheless be parallelised with high fidelity through numerical optimisation (recall Fig. 5.1(a)). Using independently optimised gates as initial guesses may therefore provide a more efficient starting point for suppressing crosstalk than random initialisation. However, because naive parallelisation performs poorly for non-local pairs, this approach may be less effective for these geometries, where more substantial modifications to the individual pulse sequences may be required.

Experimental constraints and motional dynamics

The fast-gate schemes considered in this thesis assume that all SDKs within a pulse group are delivered instantaneously, neglecting the finite repetition rates of experimental laser systems. Previous two-qubit fast-gate studies have demonstrated theoretically

that fidelities exceeding 99.99% can be achieved with repetition rates of 100 MHz to 300 MHz [46], while laser systems capable of delivering ultrafast pulses at 5 GHz have been demonstrated experimentally [79]. Although the pulse-count constraints imposed in § 5.3 reduce the sensitivity of the proposed gates to compounding SDK errors, they do not explicitly account for the temporal separation required between SDKs in a pulse group. Consequently, the experimental feasibility of the identified parallel sequences remains uncertain, particularly for pulse groups containing many SDKs. Future work should therefore incorporate finite laser repetition rates directly into the optimisation, as previously investigated for two-qubit fast gates by Gale *et al.* [65], to determine whether high-fidelity parallel operations remain achievable under realistic timing constraints. To consolidate this future study, it should also expose the parallel fast gates to realistic noise sources enabling an assessment of their robustness. In particular, previous fast gate work has identified random intensity noise and timing jitters between pulse groups to be significant noise sources [68].

To implement this in my codebase, one could extend the cost function, Eq. 5.1, used for parallel optimisation to include an additional term that directly penalises the accumulation of SDK errors, characterised by ϵ_{SDK}. A heuristic cost function for this approach is given by

$$J = (1 - \mathcal{F}) + \xi \mathcal{N} \epsilon_{\mathrm{SDK}}, \tag{6.1}$$

where $\mathcal{N}$ is the total number of SDKs and ξ controls the relative weighting of the two terms. A more detailed analysis of SDK errors that accounts for correlations between pulses would need to be implemented [67] according to the experimental characteristics of the laser system. Nonetheless, a cost function such as Eq. (6.1) could simultaneously optimise gate fidelity and SDK resources, potentially reducing the need for the iterative resource search developed in § 5.3.1. However, balancing these competing objectives would require careful selection of ξ to avoid search density limitations as imposing strong penalties on the total SDK count could prevent the optimiser from identifying high-fidelity gates.

A further limitation of this thesis is the exclusive consideration of axial motional modes. Recent theoretical work by Grosser *et al.* [66] has demonstrated that high-fidelity

fast gates can also be realised using radial motional modes, with gate durations ranging from hundreds of nanoseconds to microseconds. Unlike the axial motion considered here, radial dynamics are sensitive to micromotion arising from the RF trapping field, which must therefore be incorporated into the gate design. Radial modes also offer an additional resource for parallelisation, as demonstrated by Zhu *et al.* [69], who experimentally implemented simultaneous entangling gates using orthogonal motional modes (i.e. axial and radial). In practice this means that two gates that independently excite these orthogonal modes are inherently parallel. Extending the present framework to radial fast gates could therefore provide opportunities for suppressing crosstalk or increasing the number of simultaneous operations. In particular, the different spatial structure of radial motional modes may alter the propagation of disturbances and the resulting crosstalk, although whether this provides an advantage over axial fast gates remains to be established. As the underlying optimisation framework already accommodates radial motion and micromotion, investigating these effects represents a natural extension of the present work.

Extensions to general multi-qubit operations

Finally, although the multi-qubit operations investigated in this thesis consist of simultaneous two-qubit gates acting on disjoint ion pairs, the theoretical framework developed in Chapter 3 can be extended to more general multi-qubit entangling operations. Gu *et al.* [80] demonstrated that simultaneous two-qubit interactions involving overlapping ion pairs can generate useful multi-qubit entangling operations, while Peleg *et al.* [81] investigated the efficient design of programmable multi-qubit entangling gates in large trapped-ion systems. Such operations are particularly relevant to quantum error correction, where three-qubit gates have been proposed to reduce circuit depth and improve fault-tolerant stabiliser measurements [82]. Within the fast-gate framework considered here, extending the numerical optimisation to these operations would involve modifying the target entangling-phase matrix to allow interactions between overlapping pairs of ions. However, satisfying the additional phase constraints may increase the required gate durations and control resources, particularly when multiple ions participate in several interactions. Investigating the fidelity, resource requirements and dependence on motional

propagation timescales of these more general entangling operations would therefore provide a natural extension of this thesis. Indeed, one could perform a similar resource analysis by comparing the implementation of an algorithm using direct multi-qubit gates with its decomposition into sequences of two-qubit gates. This would establish whether the reduction in circuit depth outweighs the additional gate durations and SDK resources potentially required for direct multi-qubit operations.

6.3 Impact of parallel fast gates

Ultimately, this thesis demonstrates that high-fidelity parallel fast entangling gates are theoretically realisable, although their computational advantage depends strongly on the spatial configuration of the addressed ions and the available control resources. While inherent parallelisation is particularly effective for well-separated local pairs, numerically optimised gates enable more general parallel operations at the cost of additional constraints. These findings highlight that the collective motional dynamics enabling high-connectivity fast gates also present a central challenge to their parallelisation as was found for adiabatic gates [1]. Whilst I have demonstrated the existence of high-fidelity parallel fast gates, reducing circuit depth does not necessarily reduce the overall execution time, as increased parallel gate times may outweigh the benefits of parallelisation. Hence, to make the most of the potential speedups, it may be advantageous to employ a combination of serial gates, naive parallelisation and optimised parallel gates, selecting the most efficient protocol according to the ion-pair geometry, required fidelity and available SDK resources. Understanding and controlling these interactions will therefore be essential to realising the combined advantages of fast gates and parallel execution in future trapped-ion quantum processors.

Bibliography

[1] C. Figgatt, A. Ostrander, N. M. Linke, K. A. Landsman, D. Zhu, et al. "Parallel entangling operations on a universal ion-trap quantum computer". *Nature* 572 (2019), pp. 368–372.

[2] X. Li, X. Yin, N. Wiebe, J. Chun, G. K. Schenter, et al. "Potential quantum advantage for simulation of fluid dynamics". *Phys. Rev. Res.* 7 (2025), p. 013036.

[3] E. Troccoli and C. Federrath. "The statistics and structure of dissipation in subsonic and supersonic turbulence". *Monthly Notices of the Royal Astronomical Society* 547 (2026), stag461.

[4] J. Preskill. "Quantum Computing in the NISQ era and beyond". *Quantum* 2 (2018), p. 79.

[5] D. P. DiVincenzo. "Two-bit gates are universal for quantum computation". *Phys. Rev. A* 51 (1995), pp. 1015–1022.

[6] M. Kjaergaard, M. E. Schwartz, J. Braumüller, P. Krantz, J. I.-J. Wang, et al. "Superconducting qubits: current state of play". *Annual Review of Condensed Matter Physics* 11 (2020), pp. 369–395.

[7] L. Henriet, L. Beguin, A. Signoles, T. Lahaye, A. Browaeys, et al. "Quantum computing with neutral atoms". *Quantum* 4 (2020), p. 327.

[8] C. D. Bruzewicz, J. Chiaverini, R. McConnell, and J. M. Sage. "Trapped-ion quantum computing: Progress and challenges". *Applied Physics Reviews* 6 (2019), p. 021314.

[9] D. P. DiVincenzo. "The physical implementation of quantum computation". *Fortschritte der Physik* 48 (2000), pp. 771–783. eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1002/1521-3978%28200009%2948%3A9/11%3C771%3A%3AAID-PROP771%3E3.0.CO%3B2-E.

[10] M. Nielsen and I. Chuang. *Quantum computation and quantum information*. Cambridge University Press, 2010.

[11] V. M. Schäfer, C. J. Ballance, K. Thirumalai, L. J. Stephenson, T. G. Ballance, et al. "Fast quantum logic gates with trapped-ion qubits". *Nature* 555 (2018), pp. 75–78.

[12] A. C. Hughes, R. Srinivas, C. M. Löschnauer, H. M. Knaack, R. Matt, et al. *Trapped-ion two-qubit gates with 99.99%+ fidelity without ground-state cooling*. 2025. arXiv: 2510.17286 [quant-ph]. URL: https://arxiv.org/abs/2510.17286.

[13] P. Wang, C.-Y. Luan, M. Qiao, M. Um, J. Zhang, et al. "Single ion qubit with estimated coherence time exceeding one hour". *Nature Communications* 12 (2021), p. 233.

[14] A. Ransford, M. S. Allman, J. Arkinstall, J. P. Campora, S. F. Cooper, et al. "A 98-qubit trapped-ion quantum computer with all-to-all connectivity". *Nature* 655 (2026), pp. 81–86.

[15] F. Arute, K. Arya, R. Babbush, D. Bacon, J. C. Bardin, et al. "Quantum supremacy using a programmable superconducting processor". *Nature* 574 (2019), pp. 505–510.

[16] W.-J. Lin, H. Cho, Y. Chen, M. G. Vavilov, C. Wang, et al. "24 days-stable cnot gate on fluxonium qubits with over 99.9% fidelity". *PRX Quantum* 6 (2025), p. 010349.

[17] A. Somoroff, Q. Ficheux, R. A. Mencia, H. Xiong, R. Kuzmin, et al. "Millisecond coherence in a superconducting qubit". *Phys. Rev. Lett.* 130 (2023), p. 267001.

[18] D. Fan, G. Liu, S. Li, M. Gong, D. Wu, et al. "Calibrating quantum gates up to 52 qubits in a superconducting processor". *npj Quantum Information* 11 (2025), p. 33.

[19] D. Bluvstein, A. A. Geim, S. H. Li, S. J. Evered, J. P. Bonilla Ataides, et al. "A fault-tolerant neutral-atom architecture for universal quantum computation". *Nature* 649 (2026), pp. 39–46.

[20] R. B.-S. Tsai, X. Sun, A. L. Shaw, R. Finkelstein, and M. Endres. "Benchmarking and fidelity response theory of high-fidelity rydberg entangling gates". *PRX Quantum* 6 (2025), p. 010331.

[21] K. Barnes, P. Battaglino, B. J. Bloom, K. Cassella, R. Coxe, et al. “Assembly and coherent control of a register of nuclear spin qubits”. *Nature Communications* 13 (2022), p. 2779.

[22] S. J. Evered, D. Bluvstein, M. Kalinowski, et al. “High-fidelity parallel entangling gates on a neutral-atom quantum computer”. *Nature* 622 (2023), pp. 268–272.

[23] R. Acharya, I. Aleiner, R. Allen, T. I. Andersen, M. Ansmann, et al. “Suppressing quantum errors by scaling a surface code logical qubit”. *Nature* 614 (2023), pp. 676–681.

[24] A. Sørensen and K. Mølmer. “Quantum computation with ions in thermal motion”. *Phys. Rev. Lett.* 82 (1999), pp. 1971–1974.

[25] A. Sørensen and K. Mølmer. “Entanglement and quantum computation with ions in thermal motion”. *Phys. Rev. A* 62 (2000), p. 022311.

[26] S. A. Moses, C. H. Baldwin, M. S. Allman, R. Ancona, L. Ascarrunz, et al. “A race-track trapped-ion quantum processor”. *Phys. Rev. X* 13 (2023), p. 041052.

[27] J. Hilder, D. Pijn, O. Onishchenko, A. Stahl, M. Orth, et al. “Fault-tolerant parity readout on a shuttling-based trapped-ion quantum computer”. *Phys. Rev. X* 12 (2022), p. 011032.

[28] J. J. García-Ripoll, P. Zoller, and J. I. Cirac. “Speed optimized two-qubit gates with laser coherent control techniques for ion trap quantum computing”. *Phys. Rev. Lett.* 91 (2003), p. 157901.

[29] I. Savill-Brown, J. J. Hope, A. K. Ratcliffe, V. D. Vaidya, H. Liu, et al. “High-speed and high-connectivity two-qubit gates in long chains of trapped ions”. *Phys. Rev. Lett.* 136 (2026), p. 190802.

[30] R. P. Feynman. “Simulating physics with computers”. *International Journal of Theoretical Physics* 21 (1982), pp. 467–488.

[31] S. Lloyd. “Universal quantum simulators”. *Science* 273 (1996), pp. 1073–1078. eprint: https://www.science.org/doi/pdf/10.1126/science.273.5278.1073.

[32] I. M. Georgescu, S. Ashhab, and F. Nori. “Quantum simulation”. *Rev. Mod. Phys.* 86 (2014), pp. 153–185.

[33] R. Blatt and D. Wineland. “Entangled states of trapped atomic ions”. *Nature* 453 (2008), pp. 1008–1015.

[34] K. R. Brown, A. C. Wilson, Y. Colombe, C. Ospelkaus, A. M. Meier, et al. “Single-qubit-gate error below $\mathbf{10^{-4}}$ in a trapped ion”. *Phys. Rev. A* 84 (2011), 030303(R).

[35] M. C. Smith, A. D. Leu, K. Miyanishi, M. F. Gely, and D. M. Lucas. “Single-qubit gates with errors at the 10^{-7} level”. *Phys. Rev. Lett.* 134 (2025), p. 230601.

[36] M. A. Nielsen. “A simple formula for the average gate fidelity of a quantum dynamical operation”. *Physics Letters A* 303 (2002), pp. 249–252.

[37] E. M. Stein and R. Shakarchi. *Real analysis*. Princeton University Press, 2005.

[38] A. G. Fowler, M. Mariantoni, J. M. Martinis, and A. N. Cleland. “Surface codes: towards practical large-scale quantum computation”. *Phys. Rev. A* 86 (2012), p. 032324.

[39] A. M. Stephens. “Fault-tolerant thresholds for quantum error correction with the surface code”. *Phys. Rev. A* 89 (2014), p. 022321.

[40] C. J. Ballance, T. P. Harty, N. M. Linke, M. A. Sepiol, and D. M. Lucas. “High-fidelity quantum logic gates using trapped-ion hyperfine qubits”. *Phys. Rev. Lett.* 117 (2016), p. 060504.

[41] S. Earnshaw. “On the nature of the molecular forces which regulate the constitution of the luminiferous ether”. *Transactions of the Cambridge Philosophical Society* 7 (1842), pp. 97–112.

[42] K. R. Brown, J. Chiaverini, J. M. Sage, and H. Häffner. “Materials challenges for trapped-ion quantum computers”. *Nature Reviews Materials* 6 (2021), pp. 892–905.

[43] D. Leibfried, R. Blatt, C. Monroe, and D. Wineland. “Quantum dynamics of single trapped ions”. *Rev. Mod. Phys.* 75 (2003), pp. 281–324.

[44] J. P. Schiffer. “Phase transitions in anisotropically confined ionic crystals”. *Phys. Rev. Lett.* 70 (1993), pp. 818–821.

[45] D. G. Enzer, M. M. Schauer, J. J. Gomez, M. S. Gulley, M. H. Holzscheiter, et al. “Observation of power-law scaling for phase transitions in linear trapped ion crystals”. *Physical Review Letters* 85 (2000), pp. 2466–2469.

[46] Z. Mehdi, A. K. Ratcliffe, and J. J. Hope. "Fast entangling gates in long ion chains". *Phys. Rev. Res.* 3 (2021), p. 013026.

[47] B. B. Blinov, D. Leibfried, C. Monroe, and D. J. Wineland. "Quantum computing with trapped ion hyperfine qubits". *Quantum Information Processing* 3 (2004), pp. 45–59.

[48] J. Benhelm, G. Kirchmair, C. F. Roos, and R. Blatt. "Towards fault-tolerant quantum computing with trapped ions". *Nature Physics* 4 (2008), pp. 463–466.

[49] J. P. Gaebler, T. R. Tan, Y. Lin, Y. Wan, R. Bowler, et al. "High-fidelity universal gate set for $^{9}Be^{+}$ ion qubits". *Phys. Rev. Lett.* 117 (2016), p. 060505.

[50] J. D. Wong-Campos, S. A. Moses, K. G. Johnson, and C. Monroe. "Demonstration of two-atom entanglement with ultrafast optical pulses". *Phys. Rev. Lett.* 119 (2017), p. 230501.

[51] D. Hucul, J. E. Christensen, E. R. Hudson, and W. C. Campbell. "Spectroscopy of a synthetic trapped ion qubit". *Phys. Rev. Lett.* 119 (2017), p. 100501.

[52] J. E. Christensen, D. Hucul, W. C. Campbell, and E. R. Hudson. "High-fidelity manipulation of a qubit enabled by a manufactured nucleus". *npj Quantum Information* 6 (2020), p. 35.

[53] N. M. Linke, D. Maslov, M. Roetteler, S. Debnath, C. Figgatt, et al. "Experimental comparison of two quantum computing architectures". *Proceedings of the National Academy of Sciences* 114 (2017), pp. 3305–3310.

[54] C. Monroe, W. C. Campbell, L.-M. Duan, Z.-X. Gong, A. V. Gorshkov, et al. "Programmable quantum simulations of spin systems with trapped ions". *Rev. Mod. Phys.* 93 (2021), p. 025001.

[55] D. F. V. James. "Quantum dynamics of cold trapped ions with application to quantum computation". *Applied Physics B* 66 (1998), pp. 181–190.

[56] R. L. Taylor, C. D. B. Bentley, J. S. Pedernales, L. Lamata, E. Solano, et al. "A study on fast gates for large-scale quantum simulation with trapped ions". *Scientific Reports* 7 (2017), p. 46197.

[57] D. Wineland, C. Monroe, W. Itano, D. Leibfried, B. King, et al. “Experimental issues in coherent quantum-state manipulation of trapped atomic ions”. *Journal of Research of the National Institute of Standards and Technology* 103 (1998), p. 259.

[58] C. D. B. Bentley, R. L. Taylor, A. R. R. Carvalho, and J. J. Hope. “Stability thresholds and calculation techniques for fast entangling gates on trapped ions”. *Phys. Rev. A* 93 (2016), p. 042342.

[59] J. I. Cirac and P. Zoller. “Quantum computations with cold trapped ions”. *Phys. Rev. Lett.* 74 (1995), pp. 4091–4094.

[60] C. D. B. Bentley, A. R. R. Carvalho, and J. J. Hope. “Trapped ion scaling with pulsed fast gates”. *New Journal of Physics* 17 (2015), p. 103025.

[61] S. Saner, O. B ăz ăvan, M. Minder, P. Drmota, D. J. Webb, et al. “Breaking the entangling gate speed limit for trapped-ion qubits using a phase-stable standing wave”. *Phys. Rev. Lett.* 131 (2023), p. 220601.

[62] L.-M. Duan. “Scaling ion trap quantum computation through fast quantum gates”. *Phys. Rev. Lett.* 93 (2004), p. 100502.

[63] Z. Mehdi. “Scalable quantum computing with two-dimensional arrays of trapped ions enabled by fast gates”. MA thesis. Australian National University, 2019.

[64] A. K. Ratcliffe, L. M. Oberg, and J. J. Hope. “Micromotion-enhanced fast entangling gates for trapped-ion quantum computing”. *Phys. Rev. A* 101 (2020), p. 052332.

[65] E. P. G. Gale, Z. Mehdi, L. M. Oberg, A. K. Ratcliffe, S. A. Haine, et al. “Optimized fast gates for quantum computing with trapped ions”. *Phys. Rev. A* 101 (2020), p. 052328.

[66] P. Grosser, M. G. Galan, I. Savill-Brown, A. K. Ratcliffe, H. Liu, et al. “Radial fast entangling gates under micromotion in trapped-ion quantum computers”. *Phys. Rev. A* 114 (2026), p. 022617.

[67] Z. Mehdi, V. D. Vaidya, I. Savill-Brown, P. Grosser, A. K. Ratcliffe, et al. “Fast mixed-species quantum logic gates for trapped-ion quantum networks”. *Phys. Rev. A* 112 (2025), p. L050601.

[68] I. Savill-Brown, Z. Mehdi, A. K. Ratcliffe, V. D. Vaidya, H. Liu, et al. "Error-resilient fast entangling gates for scalable ion-trap quantum processors". *Phys. Rev. A* 113 (2026), p. 052610.

[69] Y. Zhu, A. M. Green, N. H. Nguyen, C. Huerta Alderete, E. Mossman, et al. "Pairwise-parallel entangling gates on orthogonal modes in a trapped-ion chain". *Advanced Quantum Technologies* 6 (2023).

[70] T. Olsacher, L. Postler, P. Schindler, T. Monz, P. Zoller, et al. "Scalable and parallel tweezer gates for quantum computing with long ion strings". *PRX Quantum* 1 (2020), p. 020316.

[71] L. Cheng, S.-C. Liu, L.-Y. Peng, and Q. Gong. "Crosstalk suppression of parallel gates for fault-tolerant quantum computation with trapped ions via optical tweezers". *Phys. Rev. Appl.* 22 (2024), p. 034021.

[72] D. Kielpinski, C. Monroe, and D. J. Wineland. "Architecture for a large-scale ion-trap quantum computer". *Nature* 417 (2002), pp. 709–711.

[73] J. M. Pino, J. M. Dreiling, C. Figgatt, J. P. Gaebler, S. A. Moses, et al. "Demonstration of the trapped-ion quantum ccd computer architecture". *Nature* 592 (2021), pp. 209–213.

[74] N. Grzesiak, R. Blümel, K. Wright, K. M. Beck, N. C. Pisenti, et al. "Efficient arbitrary simultaneously entangling gates on a trapped-ion quantum computer". *Nature Communications* 11 (2020), p. 2963.

[75] P. Hyllus, W. Laskowski, R. Krischek, C. Schwemmer, W. Wieczorek, et al. "Fisher information and multiparticle entanglement". *Phys. Rev. A* 85 (2012), p. 022321.

[76] Q. Wu, Y. Shi, and J. Zhang. "Continuous raman sideband cooling beyond the lamb-dicke regime in a trapped ion chain". *Phys. Rev. Res.* 5 (2023), p. 023022.

[77] L. Cheng, S.-C. Liu, G.-Z. Yao, Y.-X. Wang, L.-Y. Peng, et al. "High-quality parallel entangling gates in long mixed-species ion chains". *Phys. Rev. A* 108 (2023), p. 042603.

[78] S. Bravyi, A. W. Cross, J. M. Gambetta, D. Maslov, P. Rall, et al. “High-threshold and low-overhead fault-tolerant quantum memory”. *Nature* 627 (2024), pp. 778–782.

[79] M. Hussain, D. Heinrich, M. Guevara-Bertsch, E. Torrontegui, J. García-Ripoll, et al. “Ultraviolet laser pulses with multigigahertz repetition rate and multiwatt average power for fast trapped-ion entanglement operations”. *Phys. Rev. Appl.* 15 (2021), p. 024054.

[80] X. Gu, J. Fernández-Pendás, P. Vikstål, T. Abad, C. Warren, et al. “Fast multiqubit gates through simultaneous two-qubit gates”. *PRX Quantum* 2 (2021), p. 040348.

[81] L. Peleg, D. Schwerdt, J. Nemirovsky, Y. Shapira, N. Akerman, et al. “Fast design and scaling of multiqubit gates in large-scale trapped-ion quantum computers”. *PRX Quantum* 7 (2026), p. 033021.

[82] J. Old, S. Tasler, M. J. Hartmann, and M. Müller. “Fault-tolerant stabilizer measurements in surface codes with three-qubit gates”. *Phys. Rev. Lett.* 135 (2025), p. 240601.